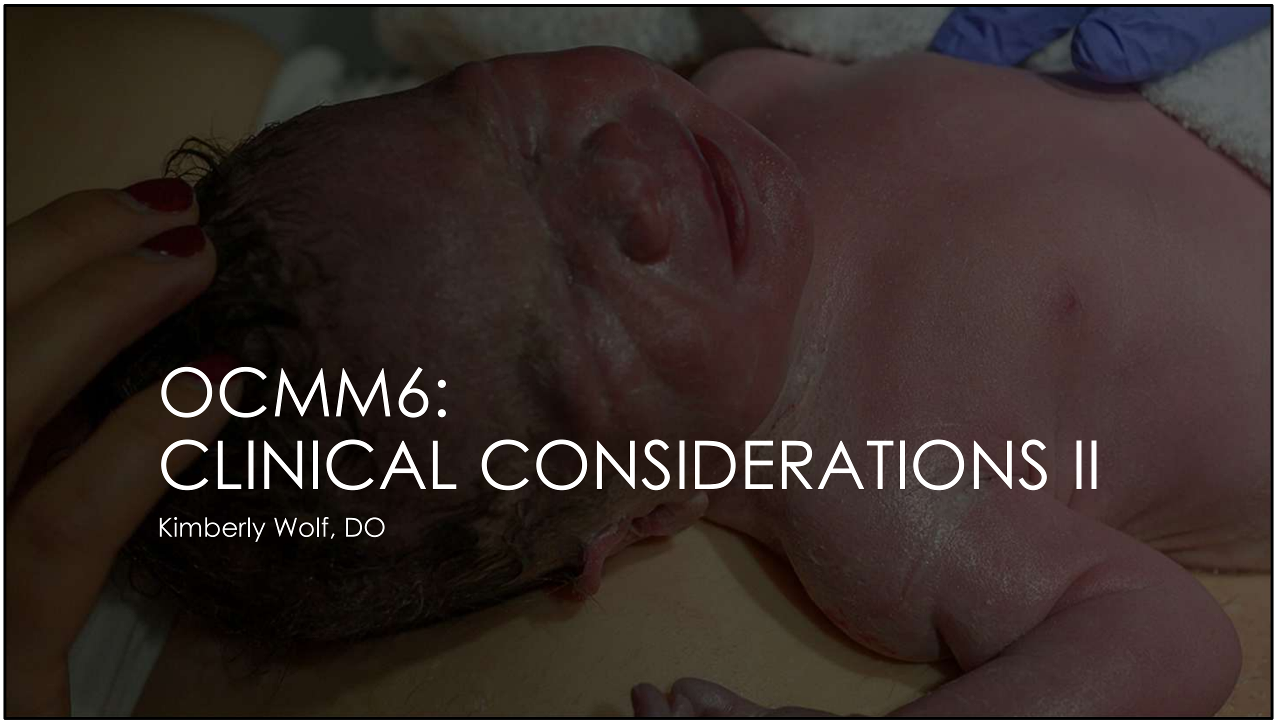


Welcome to OCMM6: Clinical considerations part 2 lecture. This is one of my fave lectures because we'll be talking about the pediatric clinical considerations.


PLAGIOCEPHALY, TORTICOLLIS,
AND FEEDING DYSFUNCTION

OBJECTIVES

1. Identify the emphasized cranial, muscular, ligamentous, & neural anatomy related to the pathophysiology of the following clinical cases:
   1. **Plagiocephaly/torticollis and feeding dysfunction**
   2. Otitis Media
   3. Concussion
2. Define the role of different treatments, including OCMM and other OMT techniques, as it relates to the pathology discussed and emphasized in the self-study
3. Compare and contrast normal motion and pathological motion of the individual cranial structures potentially associated with conditions discussed during the self-study.
4. Define relevant research as it relates to the conditions emphasized in the self-study.

CASE 1

- 2-month-old male infant presents for their well child check. Mom reports that the child is overall doing well, but that she has noticed the child's head shape looks "funny" to her. She has one other son who is 3 years old, and mom says his head never looked like this. Mom is also concerned that the child always seems to look to the right. She has tried multiple times to try to get him to look more to the left, especially while he's sleeping because she thinks that has something to do with his head shape changing. Mom first noticed this change in head shape about a month ago, but thinks it is getting worse. Mom reports that he is fussy when doing tummy time so they don't do it every day.
- Birth history: Born at 39 6/7 weeks via vaginal delivery. Mom was in labor for 42 hours, and then pushed for 3 hours. Ultimately, they used a vacuum to assist with delivery. Pregnancy was otherwise uncomplicated, and mom had good prenatal care from the first trimester. Mom and baby were both fine after delivery, and able to be discharged home together. Baby weight 8 lbs 6 oz at birth.
- Feeding history: mom has tried to breastfeed from day one, but has been very difficult. Baby had a painful latch for mom, and would really only latch on to mom's right breast. Mom has mostly been pumping and been giving her breastmilk to baby because it was so painful for her and frustrating for baby.

Let's start with a case…

CASE 1 CONTINUED

- ROS: funny-shaped head, right side preference, spits up occasionally, "colicky," feeding problems
- Meds: vitamin D
- Vaccinations: received Hep B at birth
- Allergies: NKDA
- PMHx/PSHx: no significant PMH, circumcision on DOL 1
- Trauma: none other than birth
- Family Hx: Mother and Father both alive and well with no major medical problems. One older brother who is also healthy.
- Social Hx: Lives with mom, dad, and brother. No daycare. No smoke in the home. One dog in the home.
- Developmental history: Able to briefly lift head while prone. Cooing. + social smile. Crossing midline with gaze.

CASE 1

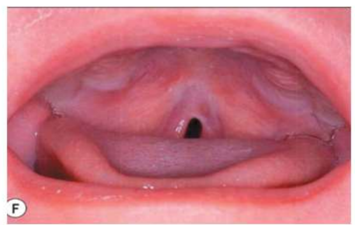
- VSS – growth appropriate
- PE:
  - Head: Plagiocephaly with flattening of right occiput, slight fullness in right maxilla and frontal bones, AFOSF, posterior fontanelle closed, right ear anterior to the left
  - EENT: PERRLA, EOMI, + red reflex, oropharynx clear, no apparent ankyloglossia, TMI, no bulging/fluid, nares patent
  - Neuro: normal tone, no focal deficits; + Moro, plantar/palmar grasp, Galant, rooting, and sucking, upgoing Babinski reflex
  - Abdomen: soft, NTND, no masses, no HSM, + BS x 4 quadrants
- OSE:
  - Cranium: left lateral strain, right condylar compression, diminished PRM, externally rotated left temporal bone
  - Cervical: rotated right throughout the cervical spine with slight hypertonicity of paraspinal musculature on the right, regionally sidebent left, hypertonic L SCM
  - Thoracic: rotated right throughout the thoracic spine with slight hypertonicity of paraspinal musculature on the right
  - Lumbar: rotated right throughout the lumbar spine with slight hypertonicity of paraspinal musculature on the right
  - Sacrum: diminished motion, S1 rotated right
  - Pelvis: restriction at the right SI joint with decreased mobility
  - UE: elevated left clavicle

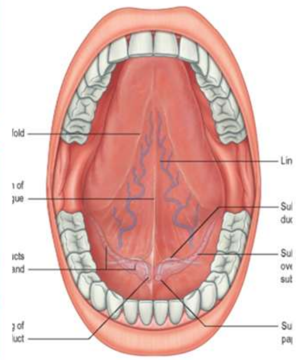
PROBLEM LIST/DDX

"Funny shaped head"	Positional Plagiocephaly Craniosynostosis
Feeding dysfunction	r/o ankyloglossia (tongue tie) or other anatomic abnormalities
Reflux	r/o anatomic abnormalities such as malrotation
Colicky/fussy	Extensive differential

DIFFERENTIAL INCLUDES:

- Cleft lip/palate
- Hypotonia
- Ankyloglossia
- Dysmorphic features – especially micrognathia
- Macroglossia
- Airway issues – malacias, stenosis, etc.





PLAGIOCEPHALY/TORTICOLLIS

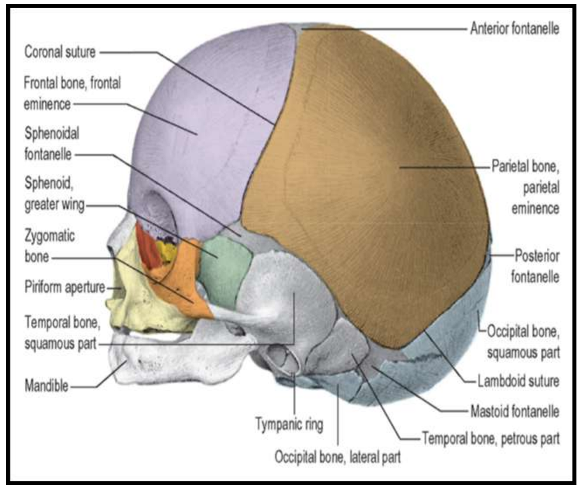
- Relevant Anatomy:
  - Newborn skull
  - Occiput
    - SBS – lateral strain
    - Condyles
      - Jugular foramen
        - CN 9-11
  - Temporal bone
    - Direct attachment with SCM
    - TMJ articulation
    - Tongue muscular attachments
    - Hyoid
  - Sternocleidomastoid
    - Sternum
    - Clavicle
    - Mastoid process of temporal bone

Newborn skull – pics ahead but seen some of these skulls in labs. Flexible and malleable because of their sutures. 30 separate bones at birth but many of those are in several pieces –occiput, temporal, sphenoid so many places where restrictions can happen where they wouldn't in an adult skull
Lateral strain – MC in plagio – will review this in upcoming lecture
Condyles – jugular foramen but also hypoglossal canal so influencing CN 12 as well
Temporal – direct SCM attachment, TMJ articulation influences feeding mechanics, tongue attached to hyoid and mandible. Hyoid extends directly from temporal bone in infants and tucked underneath mandible. Hyoid with several muscular attachments including omohyoid to scapula, sternum, etc
SCM – explains why this child's clavicle likely elevated on exam. Mastoid process of temporal bone – leads to externally rotated temporal bone on the left. Pulls medially on the mastoid process which moves the bone into ER

CRANIAL ANATOMIC CONSIDERATIONS

Neurocranium – 9 bones

- Occipital (4 parts at birth)
- 2 parietals
- 2 Frontals (2 parts at birth)
- 2 temporals (3 parts at birth)
- Sphenoid (3 parts at birth)
- Ethmoid



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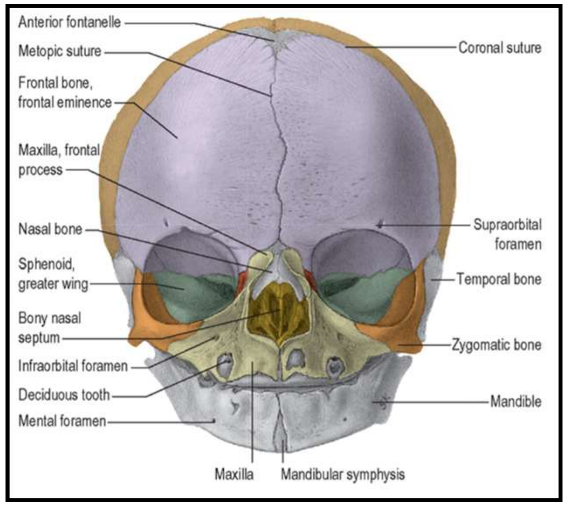
CRANIAL ANATOMIC CONSIDERATIONS

Viscerocranium/facial skeleton

- 13 bones
  - 3 nasal bones
  - 2 maxillae
  - 2 lacrimal bones
  - 2 zygoma
  - 2 palatine
  - 2 inferior nasal conchae
  - Vomer
- Other
  - Mandible
  - Hyoid
  - Middle Ear Ossicles



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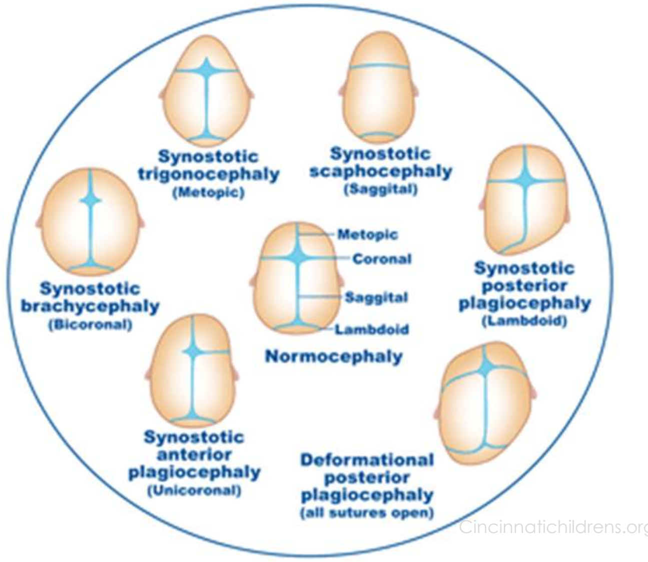
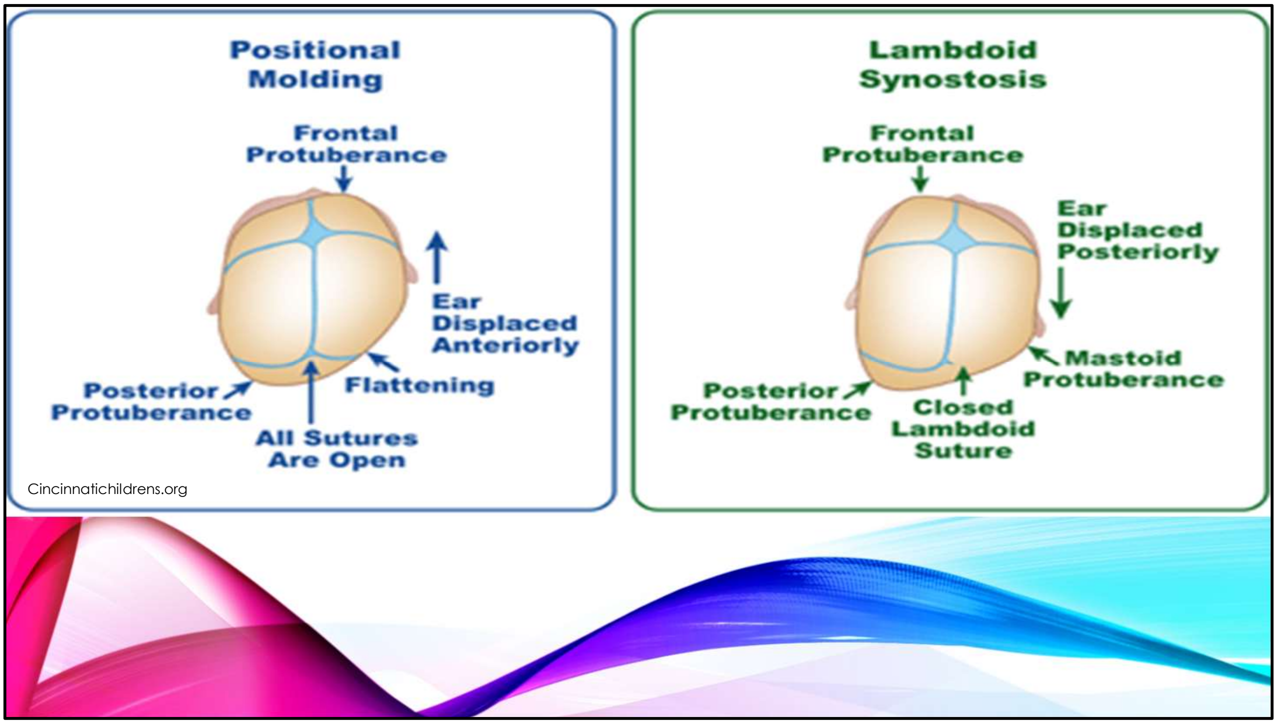
CRANIOSYNOSTOSIS

- Premature fusion of 1+ sutures
- 1: 2000-2500 births worldwide
  - Increased in uterine deformities and multiples, warfarin/valproate use, hyperthyroidism
- Restricted perpendicular growth leads to compensatory parallel growth
- Predictable deformities based on which suture fused
  - Sagittal 56%, Coronal 25%; multiple 13%



When look at plagio need to make sure that you don't miss craniosynostosis
Note the difference between deformational posterior plagiocephaly and synostotic posterior plagiocephaly



A closer look at these differences
Not ear positioning and frontal prominence as main differences

GENETICS

- Most single suture craniosynostosis is spontaneous
- Multiple sutures usually part of genetic syndrome
  - Mutations seen in fibroblast growth factor receptor family (FGFR)
  - Most are single gene mutations
  - Apert, Crouzon, Pfeiffer syndromes

Apert Syndrome

Crouzon Syndrome

POSITIONAL PLAGIOCEPHALY

- Aka deformational plagiocephaly
- Risk factors:
  - Torticollis or other conditions that decrease ROM
  - Supine sleep position
  - Decreased activity level
- Males > Females
- Risk for flattening at birth:
  - Multiples
  - Cephalohematoma
  - Primiparity
  - Long labor
  - Assisted deliveries

www.aappublications.org

Back to sleep campaign – increased risk of SIDS/SUIDS- ABCs
Minimal tummy time just like in our case
Long labor and assisted delivery just like our case as well

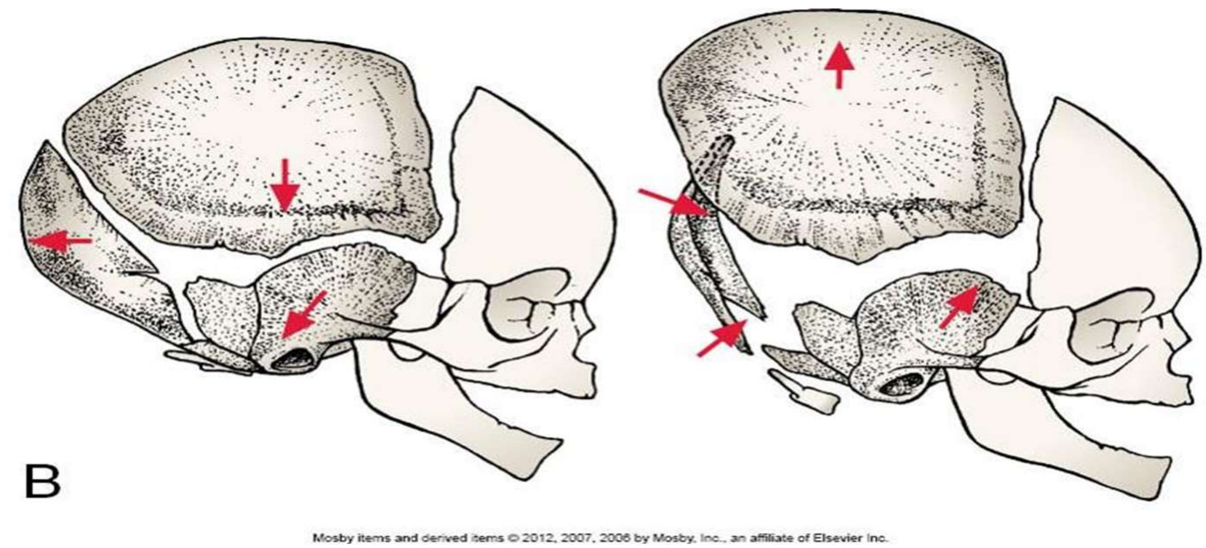
OSTEOPATHIC CONSIDERATIONS FOR THE NEWBORN

- Labor forces + anatomy → compression
- Interplay between PRM and thoracic respiration
  - Usually, primary (cranial) respiration drives secondary (thoracic) inhalation
  - First breath influences restoration of normal cranial motion
  - Crying and suckling assist
  - Compression followed by expansion that occurs in natural birth "resets" the autonomics (similar to CV4 technique)

Image: http://dou-la-la.blogspot.com/2010/08/postpartum-ocd-part-2-of-2-mom-who.html
Can be altered by C-section or vacuum assisted deliveries

Mom had 3 hours of pushing and 42 hours of labor
First breath – should be a loud cry to help
Molding seen here – "normal" but does it look comfy to you?

CRANIAL MOLDING AFTER VAGINAL BIRTH



https://classconnection.s3.amazonaws.com/372/flashcards/2357372/jpg/molding-144A3DDC6686D92DF5A.jpg


MAYBE
SHE'S
WITH
MAYBE
IT'S CRANIAL

RESEARCH

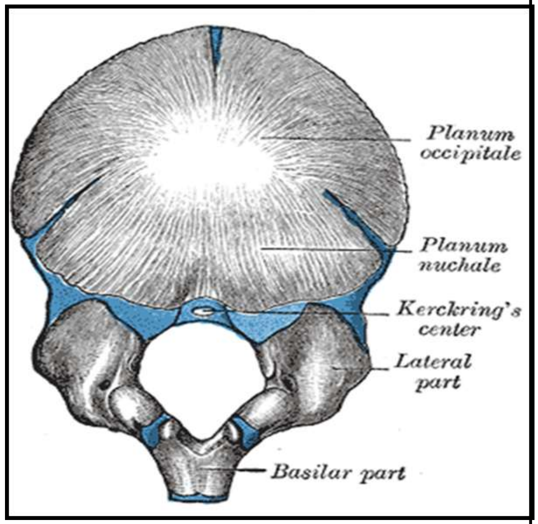
- **JOM - Relation of disturbances of craniosacral mechanisms to symptomatology of the newborn: study of 1250 infants**
    - Goal: Explore relationship between anatomic-physiologic disturbances of the craniosacral mechanism and symptoms
    - 1250 babies assessed
        - <12% had “normal” cranial mechanisms (including 6 that had a single area of restriction likely to resolve spontaneously with only breathing, crying, and sucking).
- **JOM – Osteopathic Evaluation of Somatic Dysfunction and Craniosacral Strain Pattern Among Preterm and Term Newborns**
    - Occipital bone presented with highest rate of intraosseous lesions
    - Highest rate of SD – pelvis

Frymann, V. “Relation of disturbances of craniosacral mechanisms to symptomatology of the newborn: study of 1250 infants.” *JAOA*, vol. 65, pp. 1059-1075, June 1966. Reprinted in *Clinical Cranial Osteopathy*. Ed. Richard Feeley, D.O.
Image: https://www.youtube.com/watch?v=DhSdYgskcRw

One hundred fifty-five preterm and term newborns met the study's eligibility criteria. The highest rate of somatic dysfunction was found in the pelvic area of 63 newborns (40.7%). The sacroiliac joints were compressed unilaterally or bilaterally in 82 newborns (52.9%); the lumbosacral junction was restricted in 61 newborns (39.4%), and intraosseous lesions of the sacral bone were diagnosed in 57 newborns (36.8%). The spine accounted for somatic dysfunction in 38 newborns (24.5%), with the middle thoracic and lower thoracic areas restricted in 29 (18.7%) and 21 (16.8%) newborns, respectively. Sphenobasilar synchondrosis compression and lateral-vertical strain were diagnosed in 57 newborns (36.8%), with the sagittal and the coronal sutures found restricted in 35 (22.6%) and 30 (19.4%) newborns, respectively. The occipital bone presented the highest rate of intraosseous lesions, with the left condyle compressed in 48 newborns (31%), the right condyle in 46 newborns (29.7%), and the squama in 38 newborns (24.5%).

THE OCCIPUT

- 4 parts at birth
- **Hypoglossal canal forms between the occipital condyles**
  - CN XII
- **Jugular foramen forms between temporal and occiput**
  - CN IX, X, XI
- Cartilage easily compressible
  - Birth trauma or intrauterine
  - Swelling can lead to compression

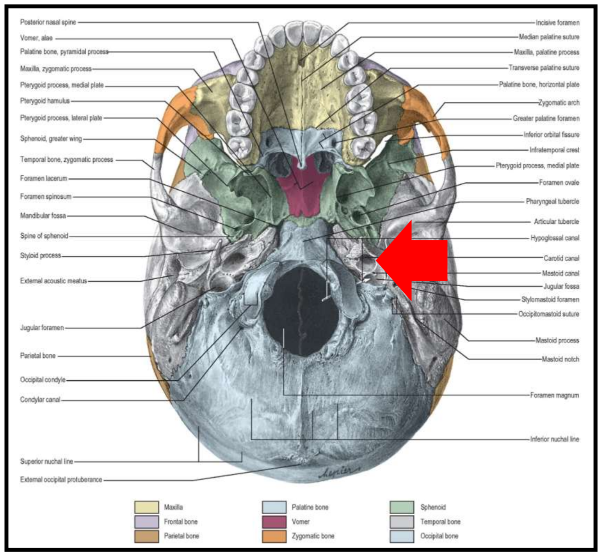


Gray's Anatomy Plate 131

Squamous portion is part of vault so allows for accommodation – molding
Anywhere you see blue is cartilage at birth

GLOSSOPHARYNGEAL NERVE IX

- Branchial motor – stylopharyngeus muscle (elevates pharynx to allow swallowing)
- Visceral motor – parasymphathetic of parotids
- Visceral sensory – from carotid sinus and body
- General sensory – from TM, upper pharynx, posterior 1/3 of the tongue
- Visceral afferent – taste from posterior 1/3 of tongue
- Exits via the jugular foramen (temporal/occiput)



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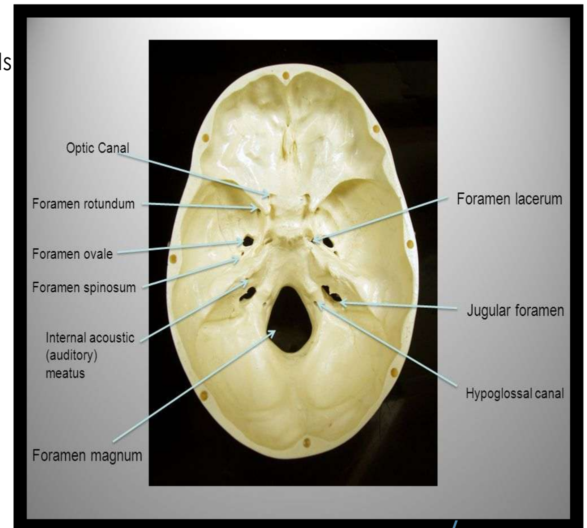
VAGUS NERVE CN X

- General visceral efferent – parasympathetic to glands of mucus membranes of pharynx, larynx, organs in neck/thorax/abdomen
- Special visceral efferent – innervates skeletal muscles of pharynx/larynx
- General somatic afferent – sensation from EAM and TM
- General visceral afferent – from thoracic and abdominal viscera, aortic body and arch
- Special visceral afferent – taste of epiglottis region of tongue
- Exits via the jugular foramen

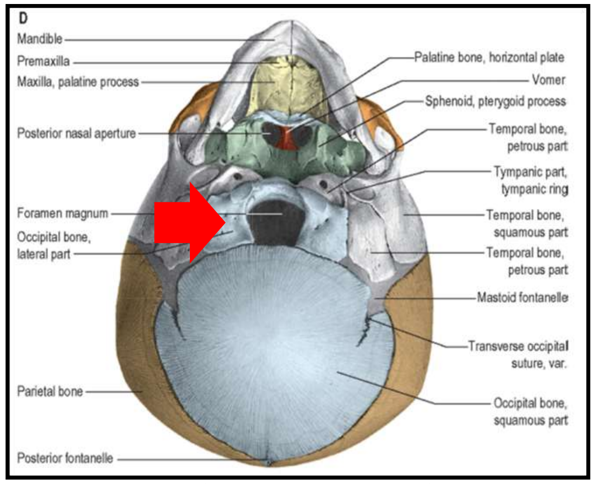


Slideplayer.com/slide/3468434/

Reaches GE sphincter – reflux
Rest of gut - colicky

SPINAL ACCESSORY CN XI & HYPOGLOSSAL CN XII

- **Spinal Accessory CN 11**
  - Motor to SCM and trapezius
  - SCM - bilateral to flex/extend head, unilaterally sidebends to same side & rotates to opposite side
  - Exits via the jugular foramen
- **Hypoglossal CN 12**
  - Motor to intrinsic tongue muscles
  - Exits via the hypoglossal canal



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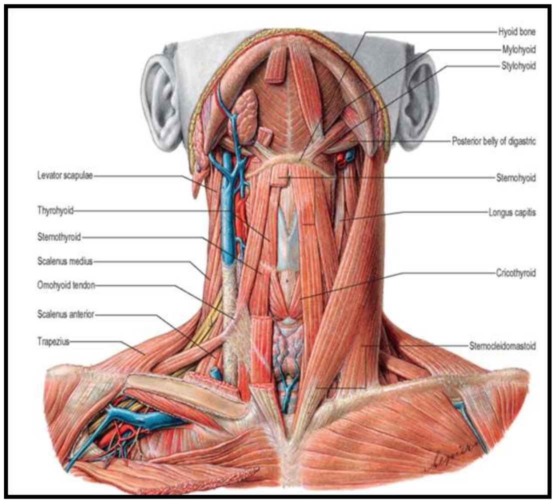
HYOID & INTRINSIC TONGUE MUSCLES

- In infants, it extends straight from temporal bones, almost inside the mandible
- In adults, more inferior (anterior to C3)

- 6 muscles attach superiorly
  - Middle pharyngeal constrictor muscle, hyoglossus, digastric, stylohyoid, geniohyoid, mylohyoid

- 3 muscles attach inferiorly
  - Thyrohyoid, omohyoid, sternohyoid



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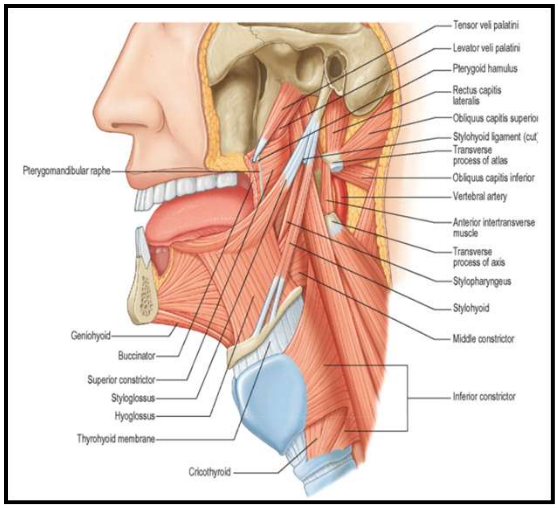
TONGUE MUSCLES

- **<u>Extrinsic muscles</u>**
  - Stabilize the tongue
  - Allow for protrusion, retraction, side to side
  - Genioglossus (mandible), hyoglossus, styloglossus, palatogossus
- **<u>Intrinsic muscles</u>**
  - Change shape, important for speech, swallowing, eating
  - Lengthen/shorten, curling, flattening/rounding
  - Superior longitudinal muscle, inferior longitudinal muscle, verticalis muscle, transversus muscle



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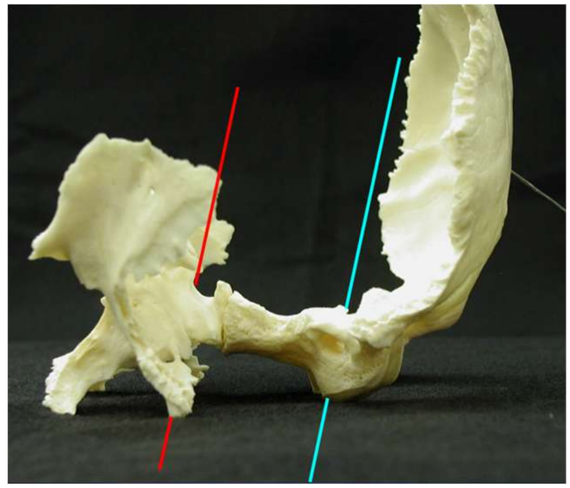
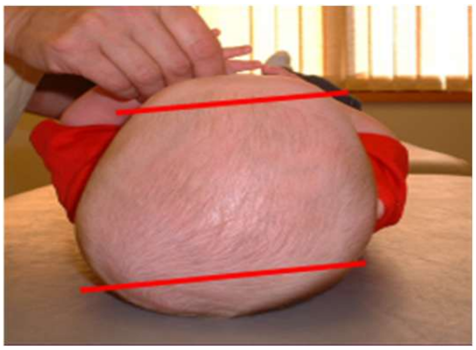
LATERAL STRAINS

- Named
  - Left or Right Lateral Strain
- Axes
  - 2 vertical axes
    - Body of the sphenoid
    - Through foramen magnum
- Rotation
  - Rotation in **same direction**
  - Shearing force at SBS causing Sphenoid and Occiput to rotate same direction on vertical axes
- Named
  - For position of base of sphenoid



LATERAL STRAINS

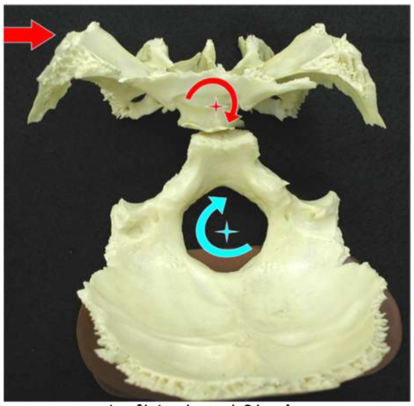
Left Lateral Strain



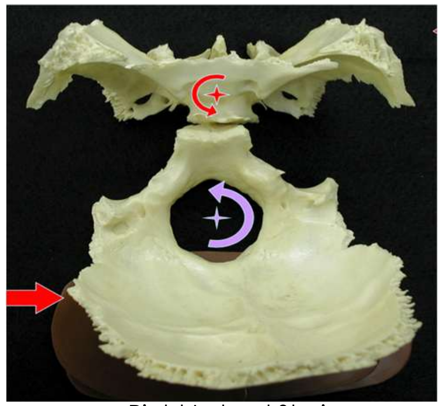
Right Lateral Strain

LATERAL STRAINS VIDEO

- http://www.youtube.com/watch?v=vZ1OOCC_DhI



Left Lateral Strain

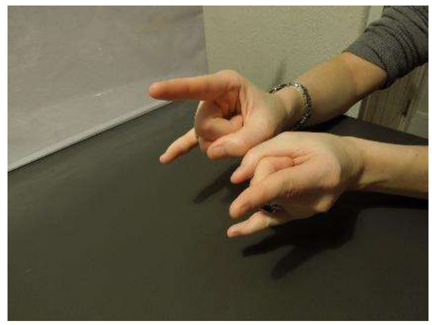


Right Lateral Strain

LATERAL STRAINS AIR HANDS

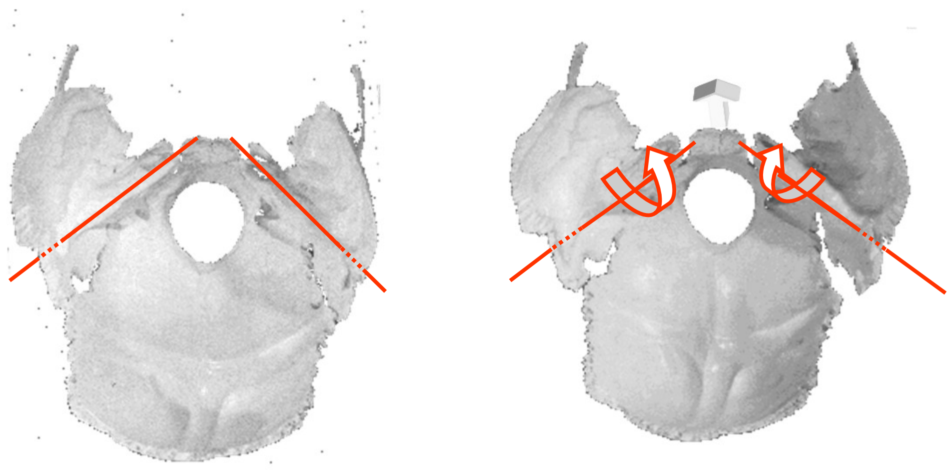
Left Lateral Strain

Right Lateral Strain



TEMPORAL BONE PRM MOTION

PAIRED BONE MOTION



OSTEOPATHIC RESEARCH

- **AAO Journal – A Pilot Study: Osteopathic Treatment of Infants with Sucking Dysfunction**
  - Only 6 patients enrolled -<6 months
  - Compared pre-post feed fat estimations in breastmilk which improved to controls following OMT
- **JOM -Osteopathic Manipulative Treatment for the Treatment of Hospitalized Premature Infants With Nipple Feeding Dysfunction**
  - Case study of premature twins who avoided placement of G-tubes after OMT treatment helped their PO nipple feeding
- **Nationwide Children's Hospital Latch study - Pilot (in process)**
  - 11 patients enrolled
  - Improvement noted in OMT vs sham groups, but no statistical significance
  - Mothers whose infants received OMT did report less pain following OMT

ADDITIONAL RESEARCH

- **Use of OMT to Treat Congenital Torticollis and Positional Plagiocephaly in an Infant: A Case Report**
  - Case study on a 7-month-old – showed improvement, 3 treatments
- **Exploring the Impact of Osteopathic Treatment on Cranial Asymmetries Associated with Nonsynostotic Plagiocephaly in Infants**
  - *Complementary Therapies in Clinical Practice 2011*
  - 12 patients, 4 treatments – showed improvement
- **Osteopathic Center for Children**
  - Retrospective chart review showed improvement – working on prospective study

ADDITIONAL RESEARCH

- **Effects of manual therapy on treatment duration and motor development in infants with severe nonsynostotic plagiocephaly: a randomised controlled pilot study**
  - 46 infants in a randomized controlled trial placed into standard treatment consisting of repositioning and orthotic treatment or to experimental group which included manual therapy
  - Treatment duration significantly shorter in experimental group compared to the control group
- **Infantile postural asymmetry and osteopathic treatment: a randomized therapeutic trial**
  - 32 infants with postural asymmetry randomly assigned to receive OMT or sham at 4 visits over course of 4 wks
  - Concluded OMT in first months of life improves degree of asymmetry in infants with postural asymmetry

https://link.springer.com/article/10.1007/s00381-016-3200-5

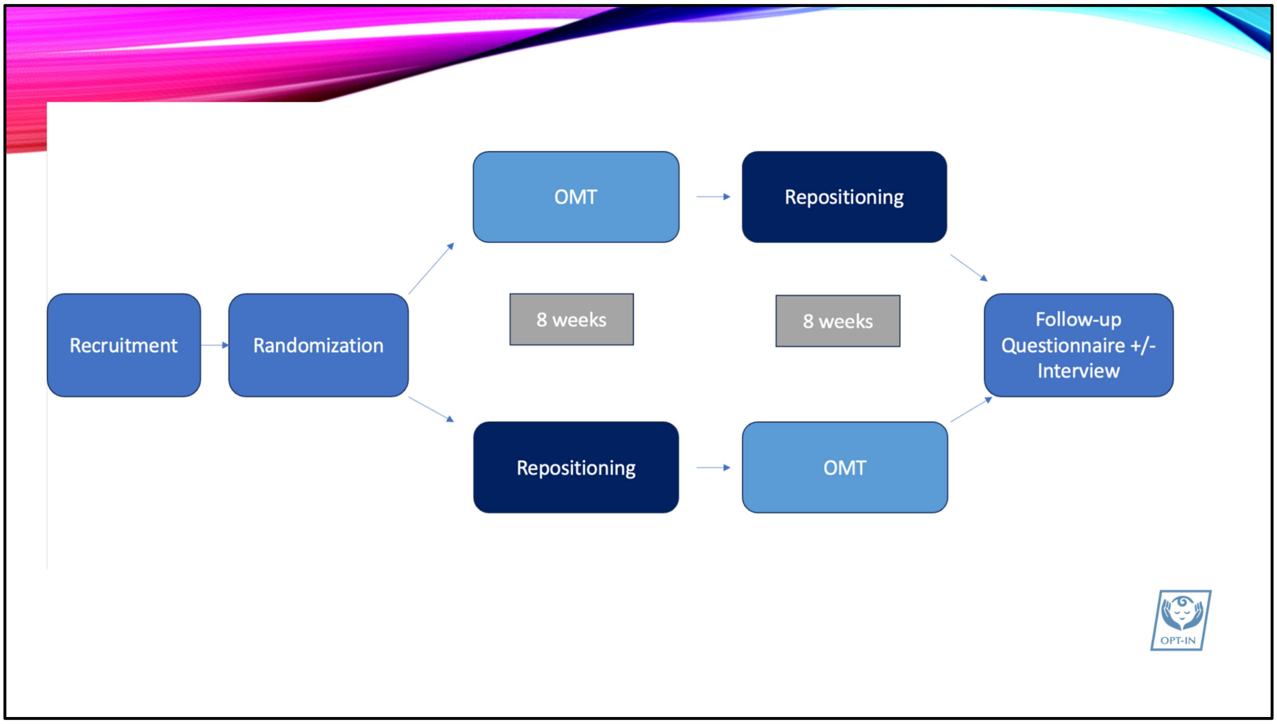
EFFECTS OF OSTEOPATHIC MANIPULATIVE TREATMENT ON CHILDREN WITH PLAGIOCEPHALY IN THE CONTEXT OF CURRENT PEDIATRIC PRACTICE: A RETROSPECTIVE CHART REVIEW STUDY
JOM, DECEMBER 2023

Results

Participants demonstrated a mean CVAI – a measure that determines the severity of DP – of 6.809 (±3.335) (Grade 3 severity) at baseline, in contrast to 3.834 (±2.842) (Grade 2 severity) after a series of OMT treatments. CVAI assessment after OMT reveals statistically significant ($p \leq 0.001$) decreases in measurements of skull asymmetry and occipital flattening. No adverse events were reported throughout the study period.

OPT-IN
An infant research study

OPT-IN: OSTEOPATHIC PLAGIOCEPHALY TREATMENT FOR INFANTS AND NEONATES



Recruitment
Randomization
OMT
Repositioning
8 weeks
8 weeks
Repositioning
OMT
Follow-up Questionnaire +/- Interview
OPT-IN

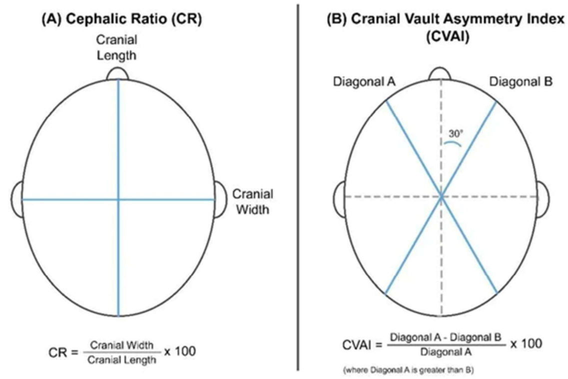
Outcomes of Interest

Primary Outcomes:

- Cranial Vault Asymmetry Index (CVAI)
- Head circumference
- Cephalic Ratio

Secondary Outcomes (evaluated at ~ 1-year):

- Developmental milestones/history
- Medical history - cases of otitis media, etc.
- Developmental delay
- Allergic rhinitis
- Vision problems (e.g., strabismus
- Torticollis
- Hospitalizations
- Use of helmet after study concluded
- Colic
- Reflux/GERD



SNEAK PEEK

	Cranial Index (CI)	Cranial Vault Asymmetry Index (CVAI)
06-13-2024	80.7 ± 0.2	+5.9 ± 0.4 Positive CVAI means the right diagonal was shorter (right flattening)

	Cranial Index (CI)	Cranial Vault Asymmetry Index (CVAI)
06-13-2024	78.3 ± 0.3	+2.8 ± 0.5 Positive CVAI means the right diagonal was shorter (right flattening)

OBJECTIVES

1. Identify the emphasized cranial, muscular, ligamentous, & neural anatomy related to the pathophysiology of the following clinical cases:
   1. Plagiocephaly/torticollis and feeding dysfunction
   2. **Otitis Media**
   3. Concussion
2. Define the role of different treatments, including OCMM and other OMT techniques, as it relates to the pathology discussed and emphasized in the self-study
3. Compare and contrast normal motion and pathological motion of the individual cranial structures potentially associated with conditions discussed during the self-study.
4. Define relevant research as it relates to the conditions emphasized in the self-study.

OTITIS MEDIA

CASE 2

- 18-month-old F presents for a sick visit with dad. Dad reports that the child has been pulling at her ear – AGAIN! Started tugging at ears again yesterday but had several ear infections over the last year. She has also had a fever of 101° F since last night. They have been giving ibuprofen for the fevers and discomfort with some relief. No ear drainage. Dad thinks this is the child's fifth ear infections total, and the most recent was one month ago. The episode last month was treated with a 10-day course of amoxicillin. Dad thinks the longest the child has gone without an ear infection has only been a couple months. He also reports that the child always seems to have a runny and stuffy nose. Has not been teething recently, and already has his 12-month molars. Have only seen their PCP about these infections. Dad is also worried that she does not hear him all the time, and feels he has to turn the TV up louder than he used to when she is watching.

CASE 2 CONTINUED

- Birth history: born at 41 weeks via C-section for failure to progress during labor. Pregnancy was otherwise uncomplicated, and baby did well after delivery.
- Feeding history: Eats a regular (picky) toddler diet. Was breastfed for first 1-2 months, but then mom went back to work and they transitioned to formula. Stopped formula at 12 months and transitioned to whole milk.
- ROS: + ear pain, fever, congestion, rhinorrhea, fussiness, ear pulling. Negative for eye redness or drainage, otorrhea, teething.
- Meds: acetaminophen and ibuprofen prn
- Vaccinations: Up to date by report
- Allergies: NKDA
- PMHx/PSHx: no significant PMH other than ear infections as per HPI, no surgeries
- Trauma: none reported
- Family Hx: Mother and Father both alive and well and both have allergic rhinitis. Has one healthy brother.
- Social Hx: Lives with mom, dad, and brother. Goes to daycare 3 days a week. Both parents smoke outdoors. One dog in the home, but it is not new.
- Developmental history: only saying four words – mama, dada, hi, bye

PE– PERTINENT POSITIVES

- Vitals: 99.6° F HR 120 RR 25 BP 90/60 BMI: 50%
- Growth charts appropriate for height, weight, and head circumference
- Gen: Alert, WDWN, fussy but consoles easily in dad's arms
- Head: NCAT, AF closed, no plagiocephaly
- Eyes: no conjunctival injection, no discharge, bilateral allergic shiners
- Ears: bilateral TM with effusions – purulent on the left, serous on the right; bulging TM on the left with erythema
- Nose: pale, boggy turbinates, clear rhinorrhea
- Mouth/throat: moist mucus membranes, no tonsillar enlargement or exudate, cobblestoning in posterior oropharynx

OSTEOPATHIC STRUCTURAL EXAM

- Head: OA restriction (FR_RS_L), diminished CRI, bogginess over bilateral maxillae without tenderness, internally rotated temporal bone on the left, compressed OM suture L>R
- Cervical spine: hypertonicity in paraspinal musculature, congestion over bilateral SCM but worse on the left, $C2FR_LS_L$
- Thoracic spine: thoracic inlet restricted-rotated right and sidebent right, T1 FR_RS_R, bogginess and hypertonicity T1-4 on the right
- Ribs: elevated first rib on the left
- Abdomen: thoracoabdominal diaphragm restriction

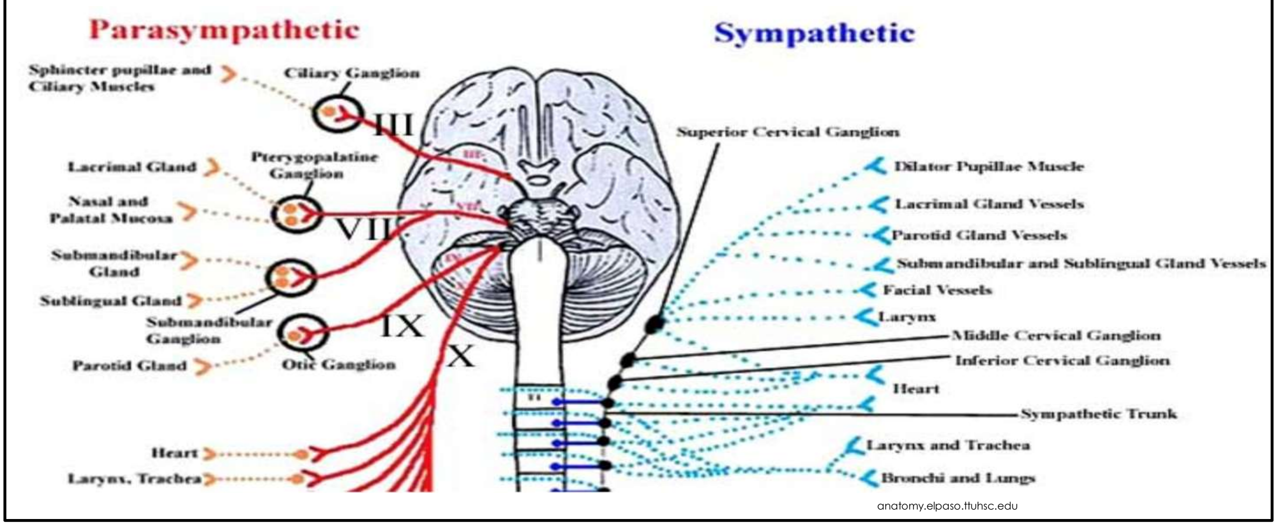
WHAT ARE YOUR DIAGNOSES?

- Allergic rhinitis
- Acute and recurrent otitis media on the left
- Serous otitis media on the right
- Concern for hearing loss
- Speech delay

HISTORY FOR OM

- Risk factors: allergies, craniofacial abnormalities, family history, formula feeding, SHS exposure
- How many episodes of OM?
- How have they been treated?
- When was the last one?
- Seen ENT?
- Otorrhea?
- Hearing changes?
- Speech delays?

ANATOMIC CONSIDERATIONS – AUTONOMICS

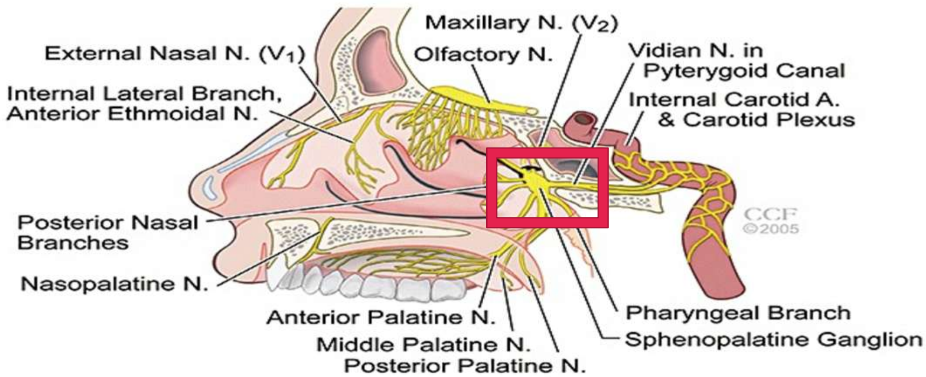


Pterygopalatine

PTERYGOPALATINE GANGLION

- Parasympathetic
  - Are some post ganglionic sympathetic fibers in the ganglion as well
  - If treated can lead to vasodilation and increased drainage, thinner secretions
- Innervated by petrosal (branch of CN7)
- Axons go to lacrimal glands, soft palate, tonsils, uvula, upper lips, gums, upper pharynx, and nasal mucosa
  - Control blood flow to the conchae

PTERYGOPALATINE GANGLION

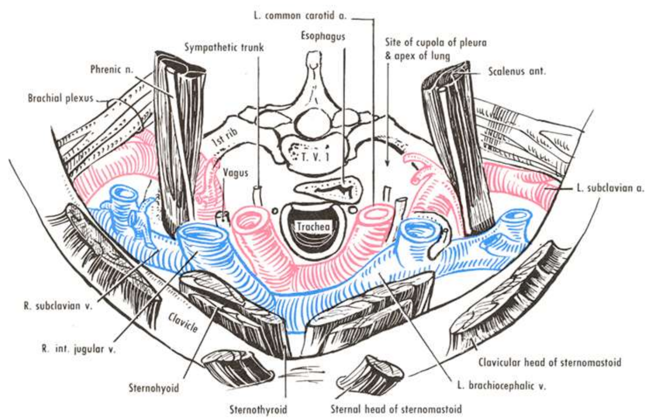


ANATOMIC CONSIDERATIONS - BIOMECHANIC

Thoracic inlet restriction

Cervical congestion

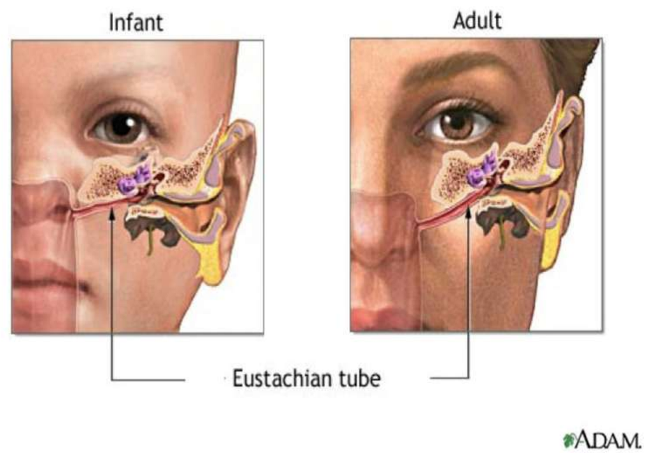
- Especially over anterior cervical chain



Embodimentetudes.com

EUSTACHIAN TUBE

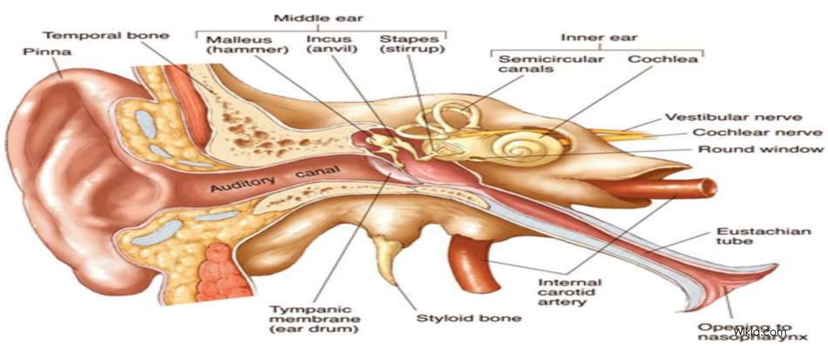
- Kids vs adults
  - More pliable and easily compressible
  - More horizontally oriented
  - Shorter
  - Poor ability to equilibrate pressure
- Changes after ~7 y/o
- Proximal 1/3 is bony, remainder is fibrocartilaginous
- Muscles: tensor veli palatini, levator veli palatini, salpingopharyngeus, tensor tympani



Adult angle is 45 degrees and kids is 10 degrees
Also feed supine, reflux, etc
10 mm infants, 36 mm adults

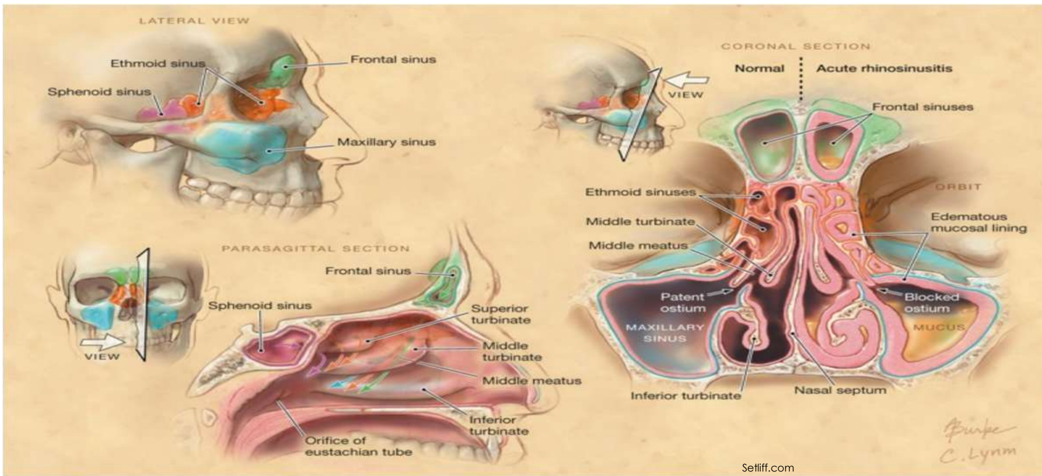
OSSEUS CONSIDERATIONS

- The middle/inner ear and bony portion Eustachian tubes lie within temporal bone
  - Squamous and petrous portions
- Distal Eustachian tubes pass through groove between petrous portion temporal bone and greater wing of sphenoid



ANATOMIC CONSIDERATIONS – OSSEUS

- Sinuses in maxillary, ethmoid, sphenoid, and frontal bones



SINUS DEVELOPMENT

Ethmoid and maxillary are present at birth but small and grow over time

Initially fluid-filled and pneumatization happens gradually

Frontal sinus does not begin to develop until ~2, but cannot be seen on x-ray until at least 5-6 years old

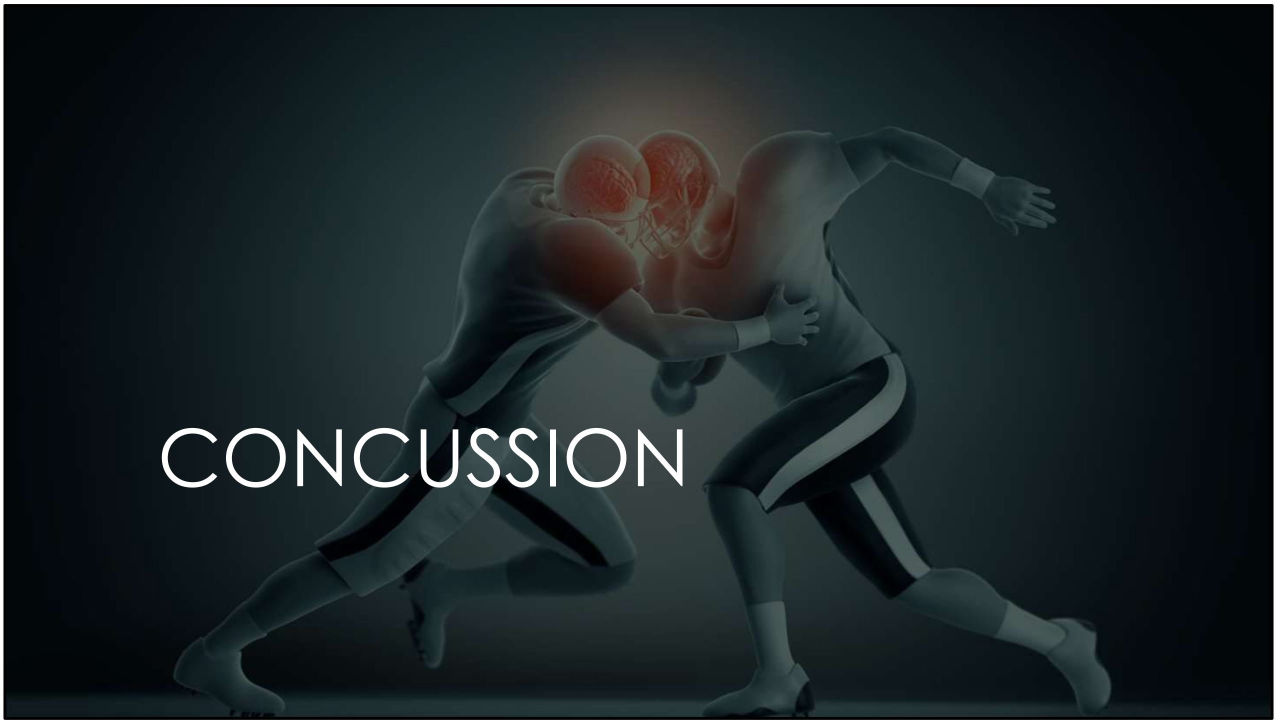
Sphenoid sinuses do not develop until adolescence

CIRCULATION/LYMPHATICS

- All about sequence!
  - Start with thoracic inlet
    - Left before right…why?
  - Then move proximal for what you would like to drain next
  - If we ultimately want to drain heads – where do you start?

OSTEOPATHIC RESEARCH

- **J Aller Ther- An Osteopathic approach to chronic sinusitis**
  - Pilot study with 16 patients
  - Reported improved pain with OMT
- **JOM – Osteopathic evaluation and manipulative treatment in reducing the morbidity of otitis media: a pilot study**
  - Only 8 patients enrolled, but most of the patients had no recurrence of symptoms
- **JOM – Effect of Osteopathic Manipulative Treatment on Middle Ear Effusion Following Acute Otitis Media in Young Children: A Pilot Study**
  - 43 patients enrolled
  - Reduced duration of effusion following AOM with OMT



Gong to cover extensive anatomy and is meant to be more of an overview – don't get bogged down by details but realize the far reaching effects that OCMM can have on these patients.   Concucsions have complicated and varied presentations – multiple sxs.
You have covered a lot of this anatomy previously including OCMM 4 and 5.

OBJECTIVES

1. Identify the emphasized cranial, muscular, ligamentous, & neural anatomy related to the pathophysiology of the following clinical cases:
   1. Plagiocephaly/torticollis and feeding dysfunction
   2. Otitis Media
   3. **Concussion**
2. Define the role of different treatments, including OCMM and other OMT techniques, as it relates to the pathology discussed and emphasized in the self-study
3. Compare and contrast normal motion and pathological motion of the individual cranial structures potentially associated with conditions discussed during the self-study.
4. Define relevant research as it relates to the conditions emphasized in the self-study.

CASE 3

- 16-year-old cisgender male presenting with headaches that started after a concussion 2 weeks ago. The headaches are constant, but do wax and wane throughout the day. They are localized to "the whole head." HA worsened by light, schoolwork, using electronics, sounds, and activity. Rest in a dark room helps the most to relieve the HA. Ibuprofen 800 mg has also given some relief, but upsets the patient's stomach so she does not take it often.
- He rates the HA currently a 5/10 which is average, but they can worsen to an 8/10 at times. Pt also has photophobia, phonophobia, decreased appetite, feels more tired yet finds it difficult to fall asleep, and per mom has also been "grumpier than normal." Pt also reports some dizziness and blurry vision bilaterally which have both been intermittent since the concussion. He also reports some neck pain that started after the injury and occasionally hears ringing in his ears.
- Trauma history: Pt suffered their first concussion 2 weeks ago while playing in a football game. Had a helmet to helmet hit. Pt does not really remember the incident but thinks that the other player hit the front of his helmet. He did briefly lose consciousness and was "out of it" for a few minutes when he woke up. His parents took him immediately to the local ER. They examined him and diagnosed him with a concussion. He was not imaged. He was discharged home with supportive care instructions including restricted activities and no football allowed.
  - Also had a broken right femur when he was 12 while playing football

CASE 3 CONTINUED

- Birth history: full-term, SVD, uncomplicated per mom
- Diet: Regular diet
- ROS: + headaches, anorexia, photophobia, phonophobia, vision changes, sleep disturbances, tinnitus, mood changes, fatigue
- Meds: 800 mg ibuprofen prn; cetirizine prn – usually in springtime
- Vaccinations: Up to date by report
- Allergies: NKDA
- PMHx/PSHx: history of allergic rhinitis; had ORIF of his right femur 4 years ago
- Family Hx: Mother and father both alive and well and have no significant history. One sister with migraines.
- Social Hx: Lives with mom, dad, and sister. Doing well in HS, but grades suffering the last couple weeks because its hard to concentrate. Wants to be an engineer when he grows up. Denies tobacco, EtOH, marijuana, or illicit drug use. Sexually active with one female partner and uses condoms.
- Developmental history: no history of developmental delays.

CASE 3

- VSS (vital signs stable)
- HEENT: EOMI, PERRLA, normal fundoscopic exam, Snellen 20/20 bilaterally
- CV/Lungs: Normal
- Neuro: CN II-XII grossly intact, DTR 2/4 b/l in UE and LE, MS 5/5 in UE and LE b/l, No atrophy, spasticity, clonus or tremors noted. Normal finger-to-nose, Romberg and heel-to-shin. Sensation and two-point discrimination intact. Proprioception intact. (normal!)
- OSE:
  - Head: SBS compression with no appreciable PRM motion, bilateral frontals externally rotated, OA FS_LR_R
  - Cervical: Tenderness/hypertonicity in suboccipital region and bilateral paraspinal cervical musculature, C3 FR_RS_R
  - Thoracic: Paraspinal muscle hypertonicity and tenderness in upper thoracic region, hypertonic trapezius worse on the right
  - Lumbar: Paraspinal muscle hypertonicity
  - Sacrum: diminished motion (PRM)

PROBLEM LIST

- Concussion
  - Headaches
  - Anorexia
  - Mood changes
  - Sleep disturbances
  - Tinnitus
  - Vision changes
  - Photophobia
  - Phonophobia
  - Neck pain
  - Fatigue
  - Difficulty concentrating (grades dropping)

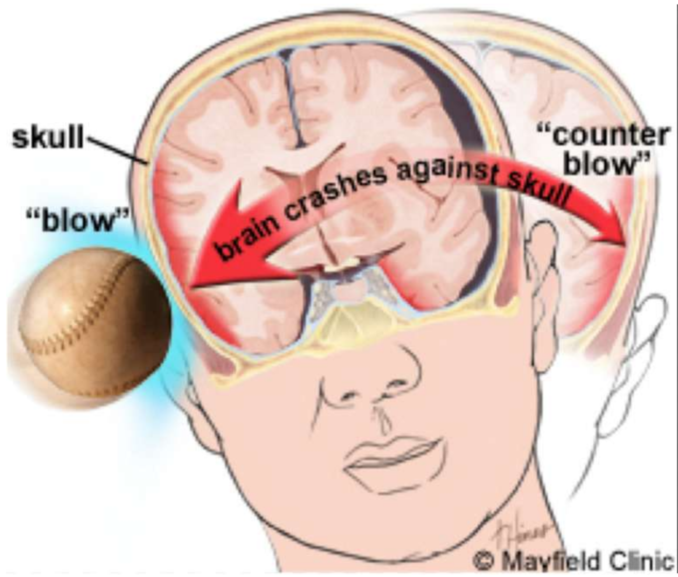
WHAT HAPPENS TO THE BRAIN WITH A CONCUSSION

Your Brain on Concussion

https://www.nytimes.com/interactive/2017/01/09/sports/football/what-happened-within-this-players-skull-football-concussions.html?ribbon-ad-idx=5&src=trending&module=Ribbon&version=context&region=Header&action=click&contentCollection=Trending&pgtype&mtrref=undefined&gwh=58D5A76521D97D95373F1038596274BB&gwt=pay&assetType=PAYWALL

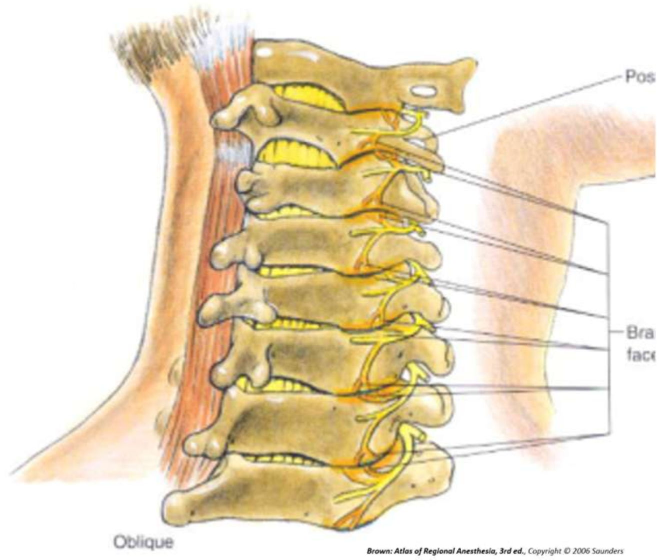
ETIOLOGY OF SYMPTOMS

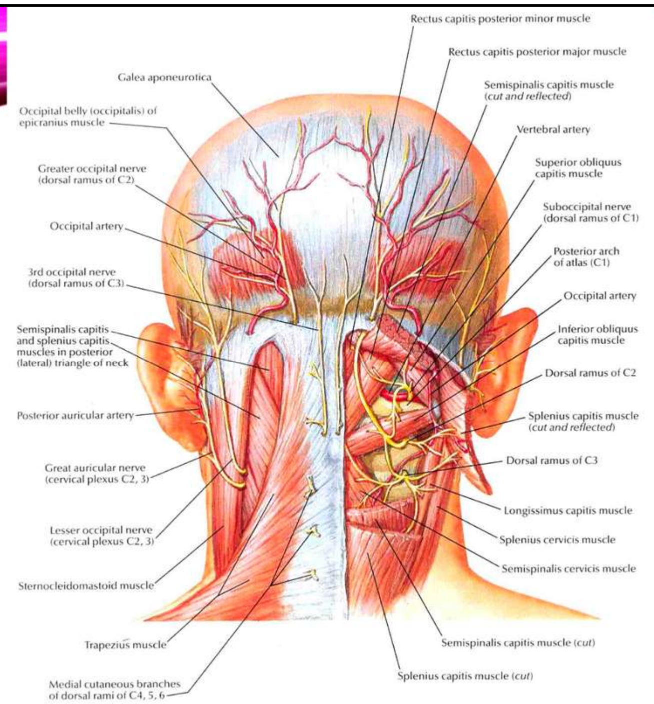
- Complex mechanisms
  - Pain from direct primary trauma
  - Inflammation and edema cause pressure on dura
    - Trigger intracranial mechanical pain generators
  - Secondary (indirect) trauma from inertia
    - Cervical spine
    - Sacrum



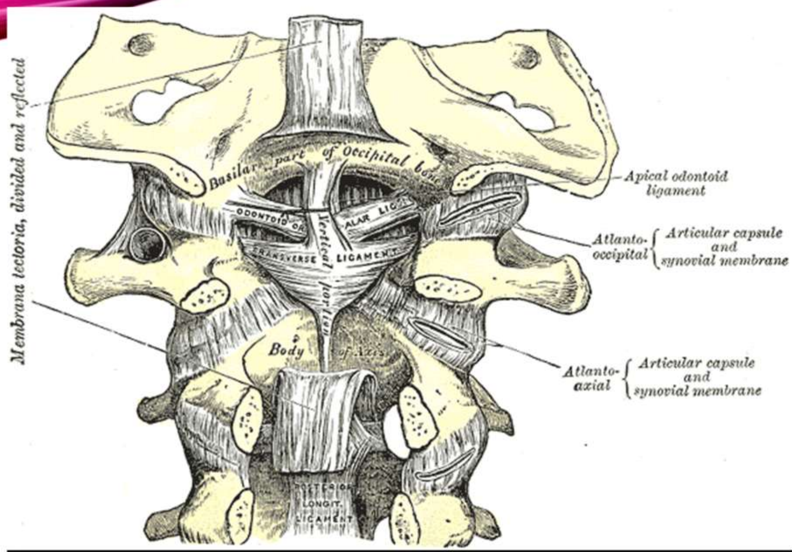
CERVICAL SPINE

- Pain generators
  - Myofascial tissue
  - Nerve compression or discogenic impingement
  - Ligamentous strains
  - Facet joints –innervated by medial branch of dorsal ramus of spinal nerve
    - Each joint supplied by its own level, and the one above
    - Pain referred to head, cervical spine, thoracic spine – reproduced from extension and rotation



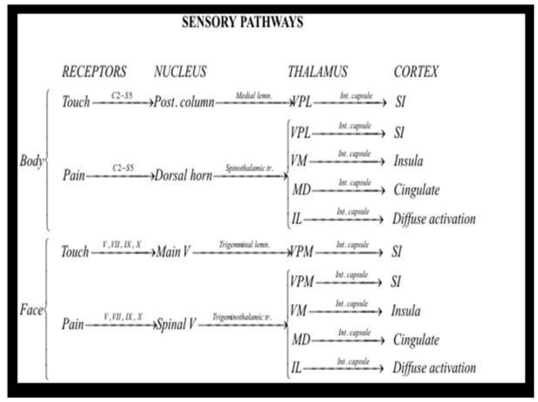


Netterimages.com



Boneandspine.com

CNS FACILITATION



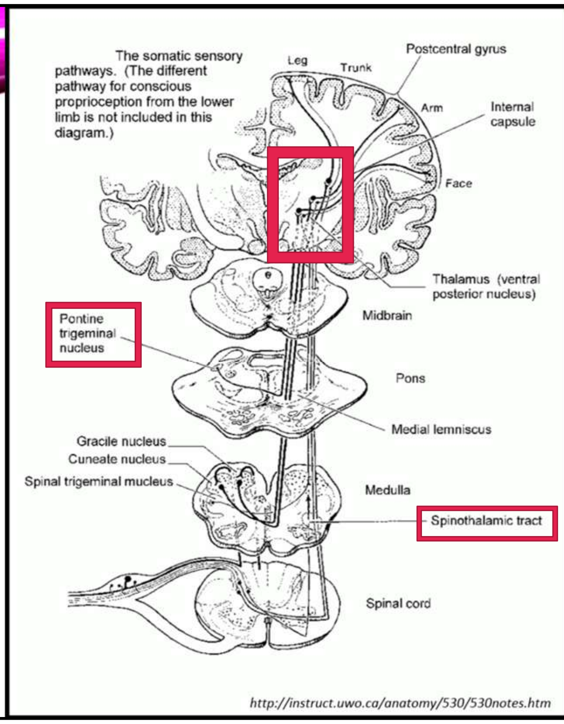
Wikipedia.org

Nociception from soft tissues in upper cervical spine carried through the anterolateral system or spinothalamic tract

- Anterior portion carries information about crude touch/pressure
- Lateral portion carries information about pain and temperature

ALS and trigeminal overlap

- Cervical strains → headaches and associated symptoms


The somatic sensory pathways. (The different pathway for conscious proprioception from the lower limb is not included in this diagram.)
Leg
Trunk
Postcentral gyrus
Arm
Internal capsule
Face
Thalamus (ventral posterior nucleus)
Midbrain
Pontine trigeminal nucleus
Pons
Medial lemniscus
Gracile nucleus
Cuneate nucleus
Spinal trigeminal mucleus
Medulla
Spinothalamic tract
Spinal cord
http://instruct.uwo.ca/anatomy/530/530notes.htm

DIZZINESS/BALANCE ISSUES

Cervical spine dysfunction → altered input to cerebellum

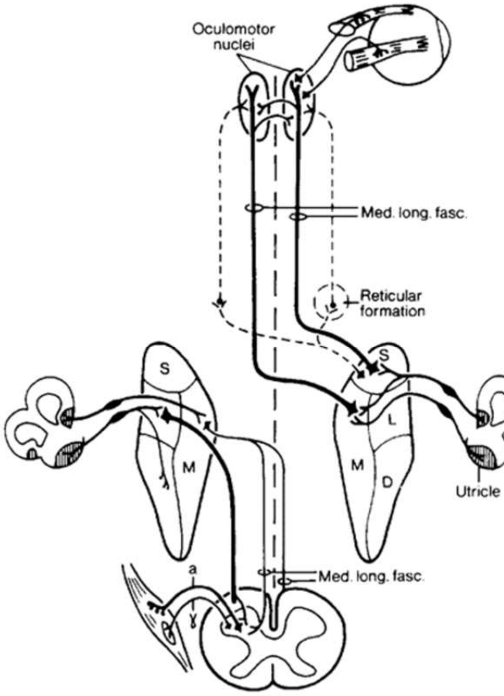
- Postural reflexes (cervico-ocular and vestibulo-ocular reflexes) play a role as well → altered motor output

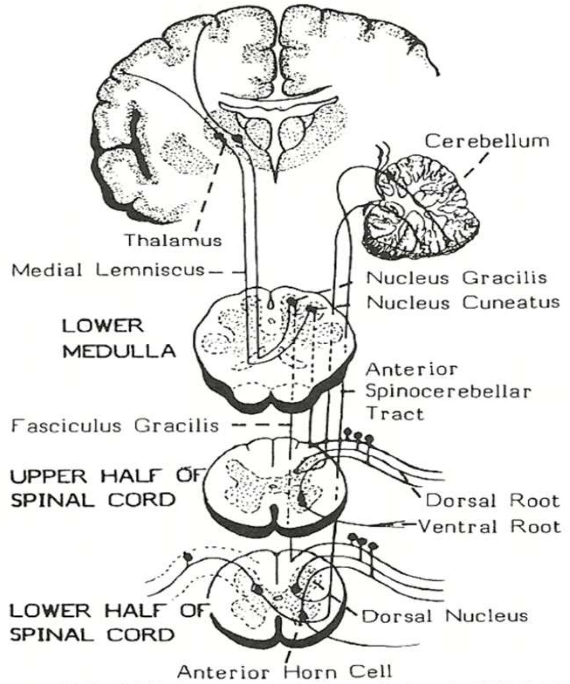
Cranial base dysfunction → Temporal bone motion altered

- Otoliths disturbed (tinnitus)
- Postural reflexes
  - Vestibulospinal reflex, vestibulo-ocular reflex, vestibulocolic reflex

Vestibular changes

- Acceleration/deceleration from trauma may cause mild prolonged "motion sickness" due to changes in sensory cells of membranous labyrinth





Hain, Timothy C. "Cervicogenic causes of vertigo." *Current opinion in neurology* 28.1 (2015): 69-73.

HEARING AND VISION PATHWAYS

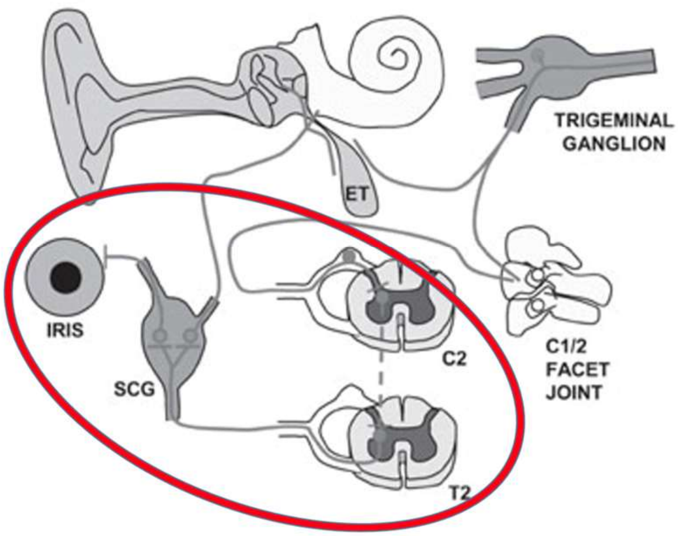
Upper cervical dysfunction → facilitation → SNS activates superior cervical ganglion →

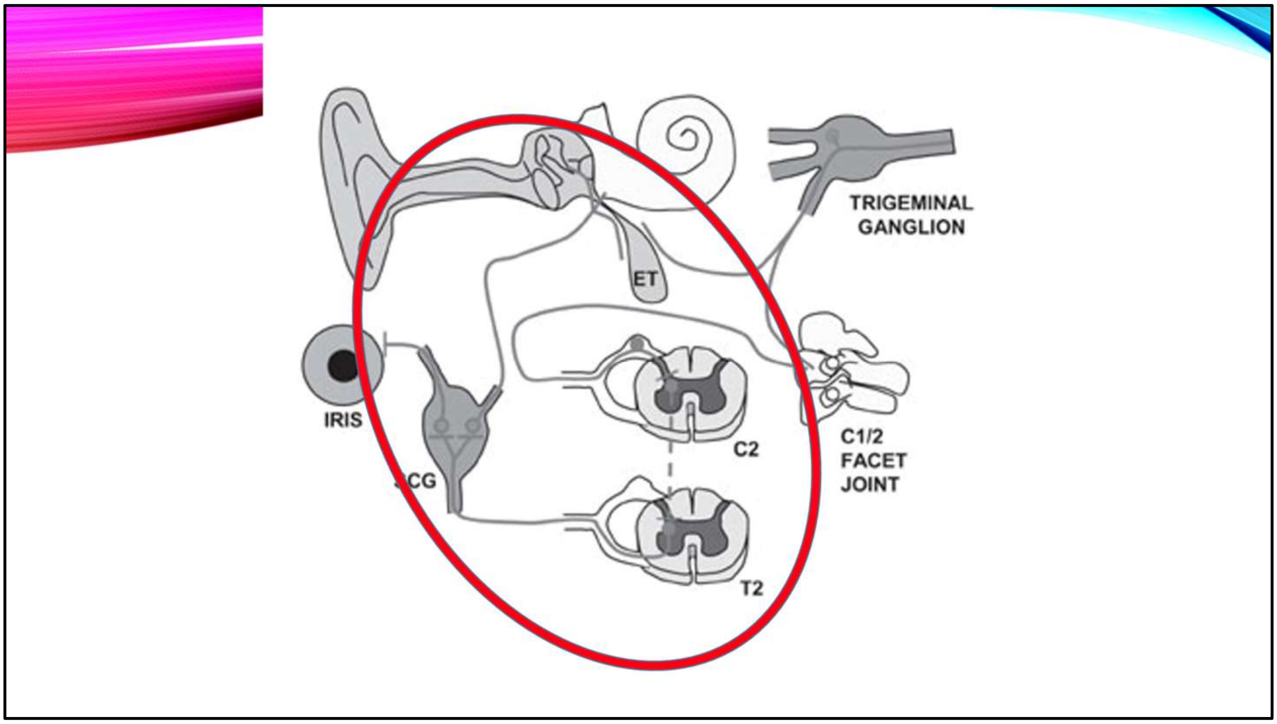
- Iris → Altered accommodation
- Middle ear → ⬇Blood flow→ effusions, tinnitus, hearing loss

Upper cervical dysfunction → facilitation of CN5 →

- Tensor veli palatini → Eustachian Tube Dysfunction
- Tensor tympani inhibiting accommodation to loud noise → phonophobia

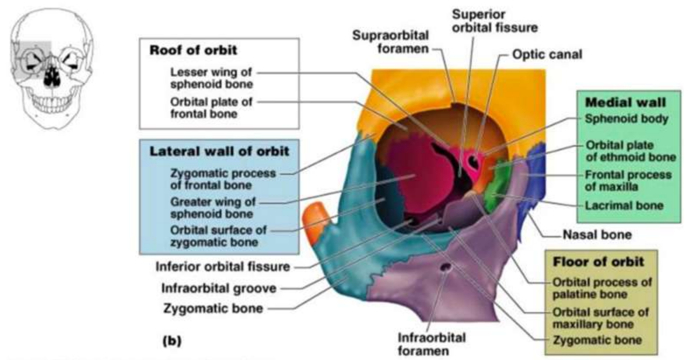
Decreased blood flow is from vertebral to cochlear artery and arterial constriction

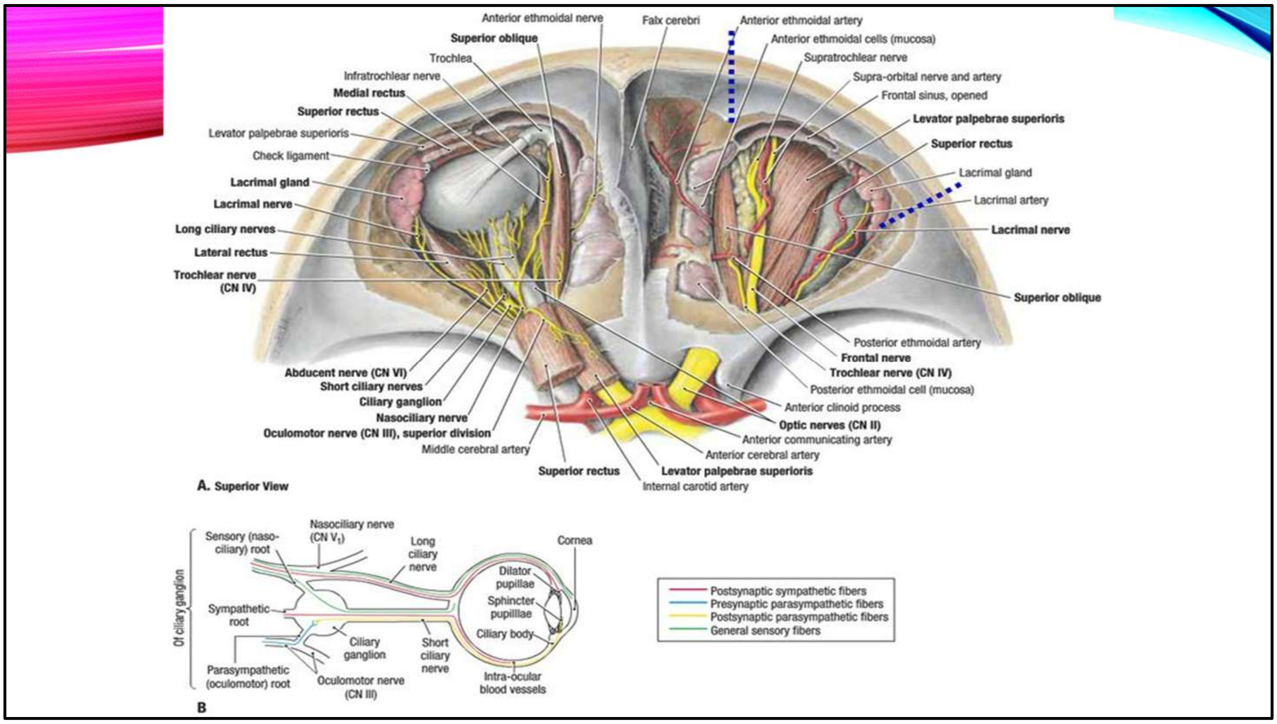

TRIGEMINAL
GANGLION
ET
IRIS
SCG
C2
C1/2
FACET
JOINT
T2


TRIGEMINAL
GANGLION
ET
IRIS
C2
C1/2
FACET
JOINT
T2

THE ORBIT

- Frontal bone
- Sphenoid
- Ethmoid
- Lacrimal bone
- Palatine
- Maxilla
- Zygoma

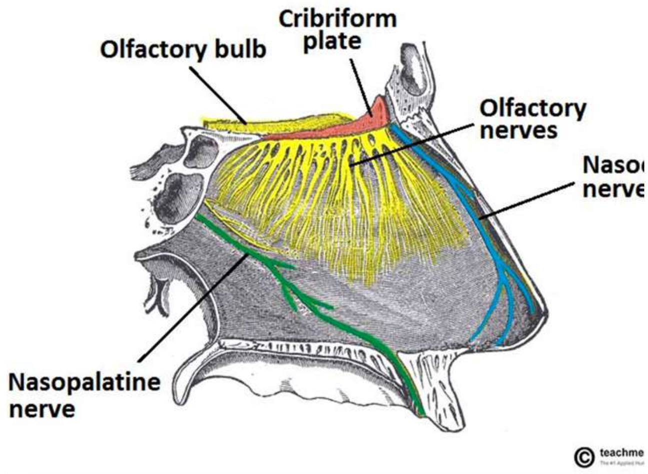


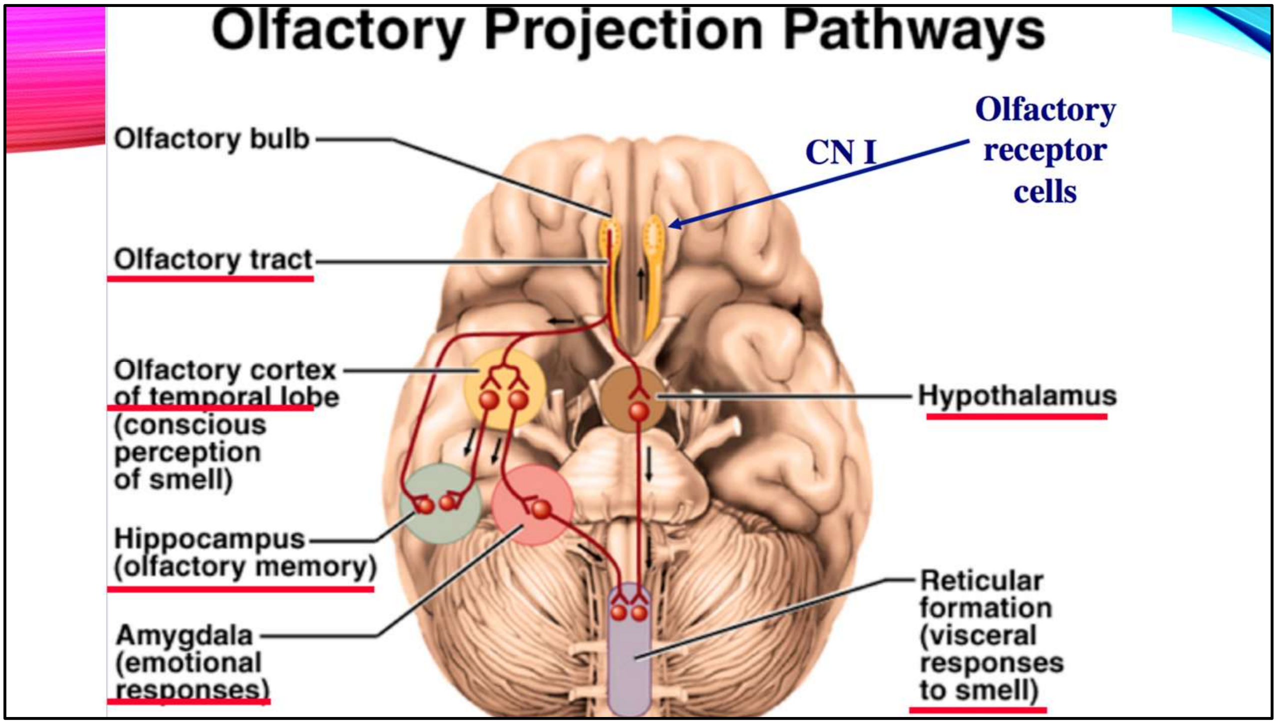


Anterior ethmoidal nerve
Falx cerebri
Anterior ethmoidal artery
Superior oblique
Anterior ethmoidal cells (mucosa)
Trochlea
Supratrochlear nerve
Infratrochlear nerve
Supra-orbital nerve and artery
Medial rectus
Frontal sinus, opened
Superior rectus
Levator palpebrae superioris
Levator palpebrae superioris
Superior rectus
Check ligament
Lacrimal gland
Lacrimal gland
Lacrimal artery
Lacrimal nerve
Long ciliary nerves
Lacrimal nerve
Lateral rectus
Trochlear nerve (CN IV)
Superior oblique
Posterior ethmoidal artery
Frontal nerve
Abducent nerve (CN VI)
Trochlear nerve (CN IV)
Short ciliary nerves
Posterior ethmoidal cell (mucosa)
Ciliary ganglion
Anterior clinoid process
Nasociliary nerve
Optic nerves (CN II)
Oculomotor nerve (CN III), superior division
Anterior communicating artery
Middle cerebral artery
Anterior cerebral artery
Superior rectus
Levator palpebrae superioris
Internal carotid artery
A. Superior View
Nasociliary nerve (CN V1)
Sensory (naso-ciliary) root
Long ciliary nerve
Cornea
Dilator pupillae
Sphincter pupillae
Ciliary body
Of ciliary ganglion
Sympathetic root
Ciliary ganglion
Short ciliary nerve
Parasympathetic (oculomotor) root
Oculomotor nerve (CN III)
Intra-ocular blood vessels
Postsynaptic sympathetic fibers
Presynaptic parasympathetic fibers
Postsynaptic parasympathetic fibers
General sensory fibers
B

SMELL ALTERATION

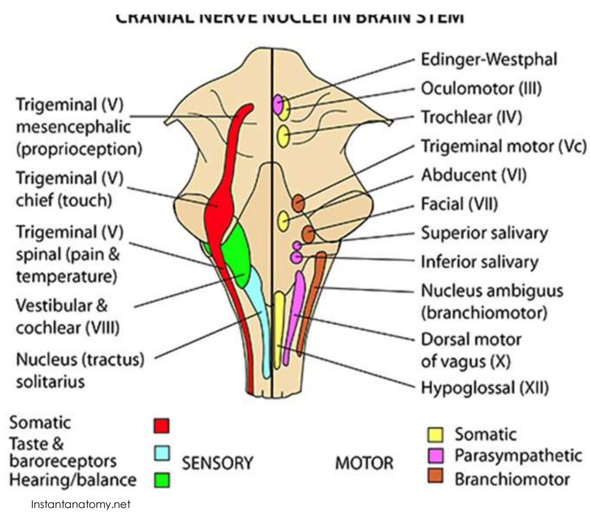
- Anosmia – partial or complete
  - Affects taste
  - Can be due fracture of cribiform plate (ethmoid)
  - Occipital or parietal trauma can also damage olfactory bulb tracts

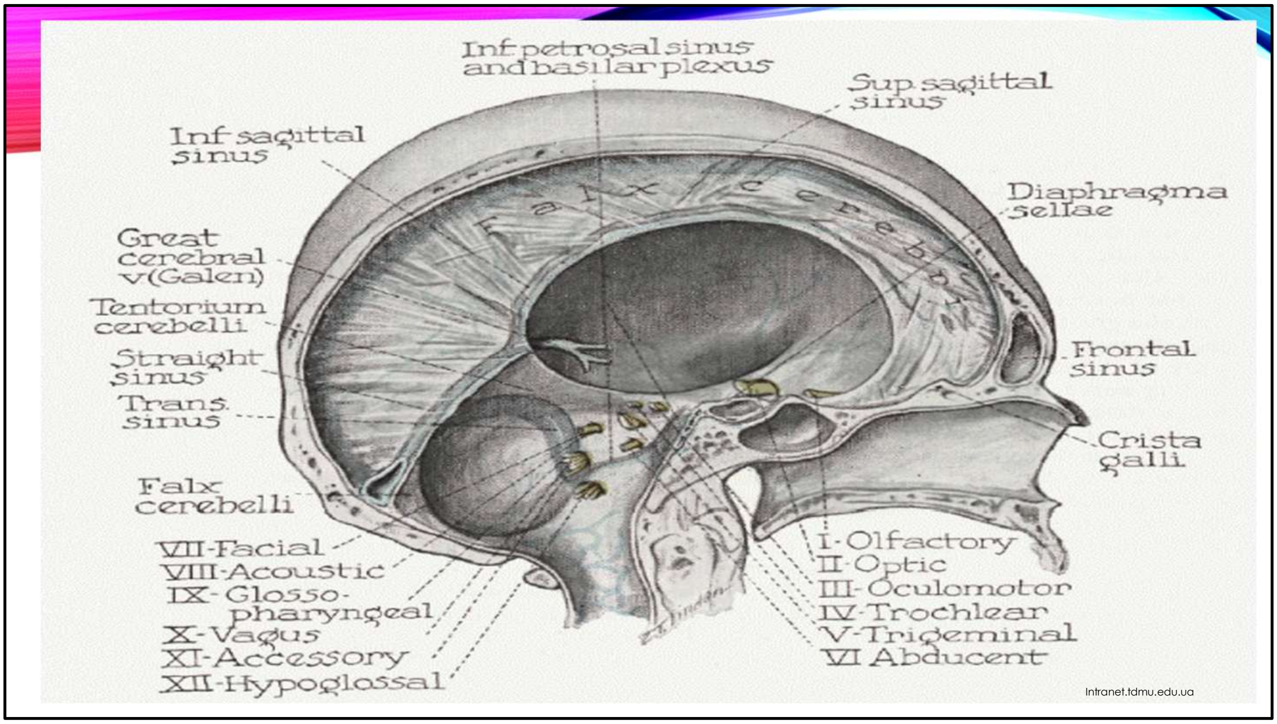


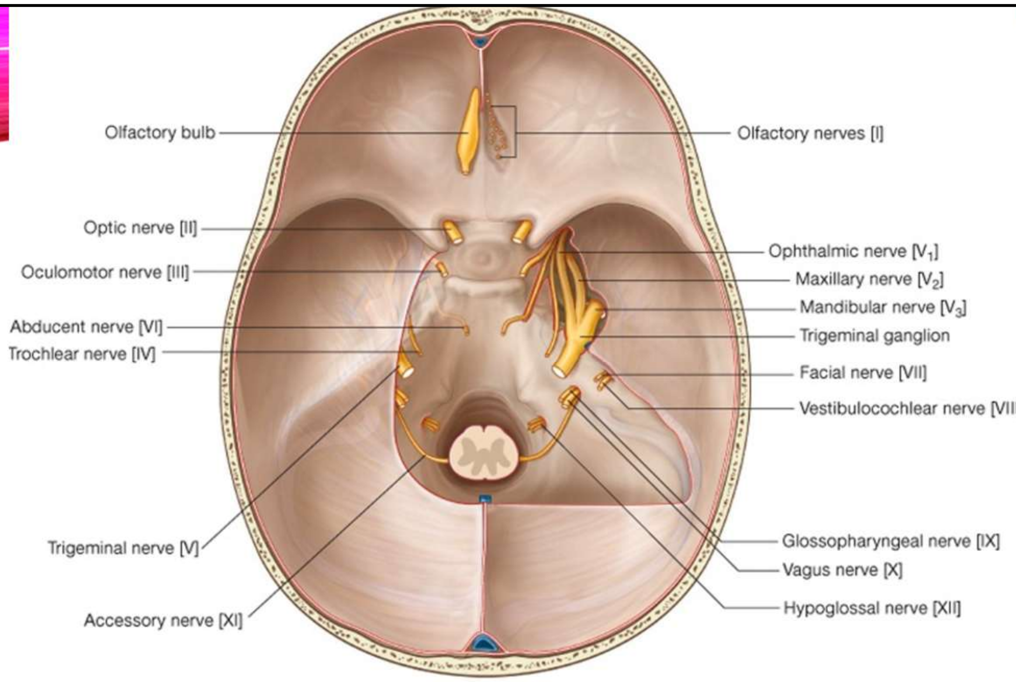

Olfactory Projection Pathways
Olfactory bulb
Olfactory tract
Olfactory cortex of temporal lobe (conscious perception of smell)
Hippocampus (olfactory memory)
Amygdala (emotional responses)
CN I
Olfactory receptor cells
Hypothalamus
Reticular formation (visceral responses to smell)

FACILITATION OF CN

- Facial Nerve: facial pain
- Glossopharyngeal and vagus: nausea
- Vagus: dysphagia and GI disturbance, appetite changes
- Spinal accessory: Neck stiffness and pain



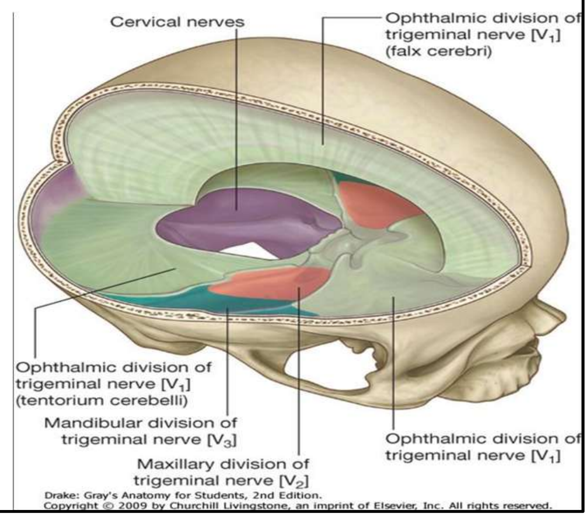

Inf. petrosal sinus and basilar plexus
Sup. sagittal sinus
Inf. sagittal sinus
Diaphragma sellae
Great cerebral v. (Galen)
Tentorium cerebelli
Straight sinus
Frontal sinus
Trans. sinus
Crista galli
Falx cerebelli
VII-Facial
VIII-Acoustic
IX-Glosso-pharyngeal
X-Vagus
XI-Accessory
XII-Hypoglossal
I-Olfactory
II-Optic
III-Oculomotor
IV-Trochlear
V-Trigeminal
VI Abducent
Intranet.tdmu.edu.ua

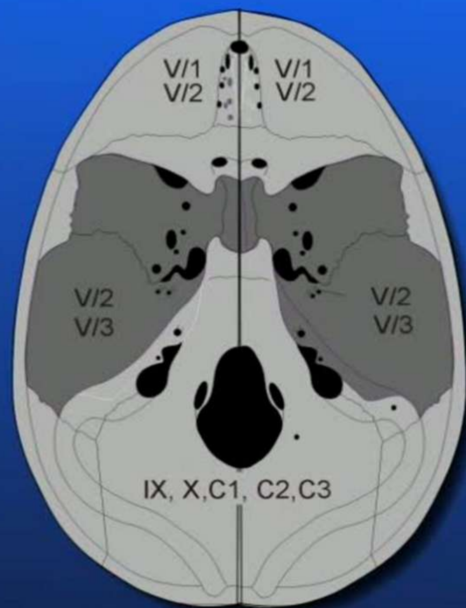


Drake: Gray's Anatomy for Students, 2nd Edition.

INNERVATION OF DURA

- No true pain fibers in brain parenchyma
- Meningeal pain fibers
  - Anterior and middle cranial fossa from CN5
  - Posterior cranial fossa: C1-3 via CN9 &10





http://loadmedical.com/craniosacral-osteopathy-duramater-vault-bones-

SO WHAT?

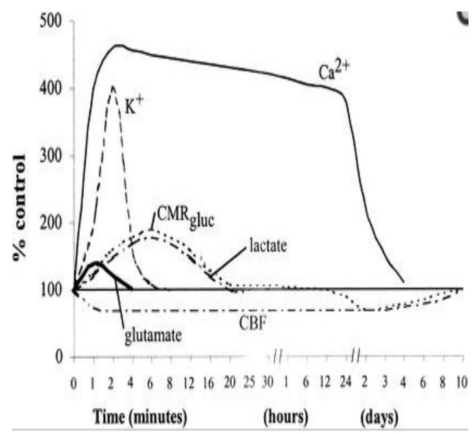
- These intimate relationships explain how even slight disruptions in the system can result in the wide variety of symptoms
  - Nausea, vomiting – CN X
  - Dizziness – CN VIII, X
  - Tinnitus – CN VIII
  - Decreased appetite – CN X

OTHER THOUGHTS ON DURA

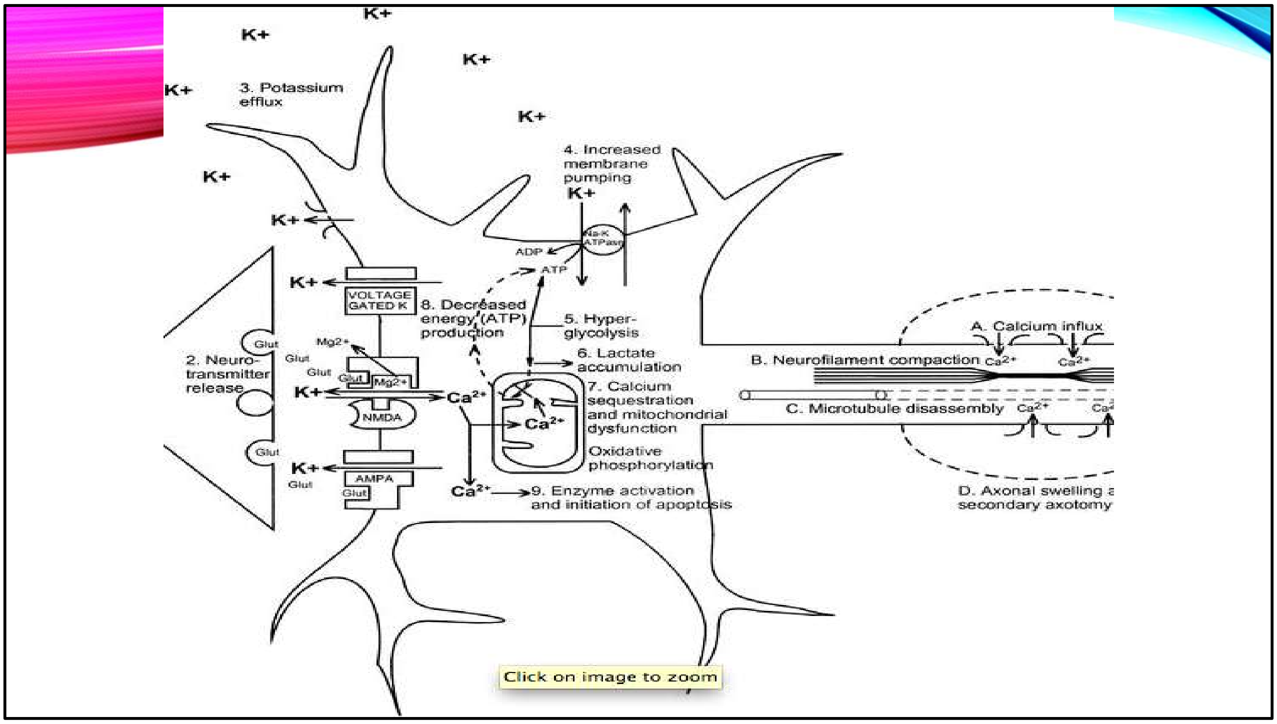
- These attachments outside of the head, help explain why the trauma to the head may lead to pain elsewhere in the body
  - Neck pain
  - Back pain
  - Sacral/pelvic pain
  - And ultimately further impaired functioning

ANYTHING ELSE?

- Studies have shown there is a pathophysiologic cascade following concussion
  - Neuronal depolarization
  - Release of excitatory neurotransmitters (glutamate)
  - Ionic shifts (K, Ca, Na, Mg)
  - Changes in glucose metabolism (to meet demands of shifts)
  - Altered cerebral blood flow (energy crisis!)
  - Impaired axonal function (poor neural connectivity)
  - Altered mitochondrial oxidative metabolism, increased free radical, increased inflammatory response
  - Increased lactic acid



Giza, Christopher C., and David A. Hovda. "The New Neurometabolic Cascade of Concussion." *Neurosurgery*. Print.



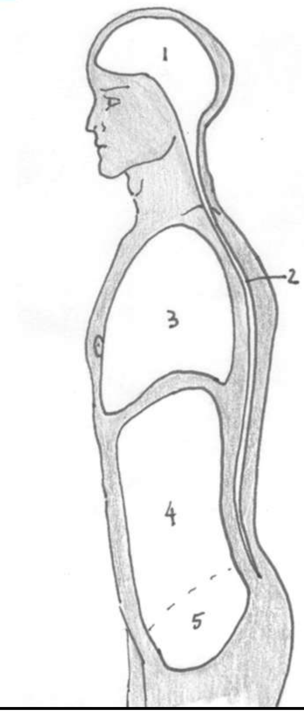
3. Potassium efflux
4. Increased membrane pumping
Na-K ATPase
ADP
ATP
VOLTAGE GATED K
8. Decreased energy (ATP) production
5. Hyper-glycolysis
6. Lactate accumulation
2. Neuro-transmitter release
Glut
Mg2+
NMDA
AMPA
7. Calcium sequestration and mitochondrial dysfunction
Oxidative phosphorylation
9. Enzyme activation and initiation of apoptosis
A. Calcium influx
B. Neurofilament compaction
C. Microtubule disassembly
Click on image to zoom

LYMPHATIC INFLUENCE

This inflammatory cascade could explain cognitive, mood, and sleep symptoms

Address the “baffles” of the body to help engage flow

- Tentorium cerebelli (and the rest of the cranial dura)
- Thoracic inlet
- Abdominal diaphragm
- Pelvic diaphragm



Respiratory circulatory model of osteopathy

GLYMPHATIC SYSTEM

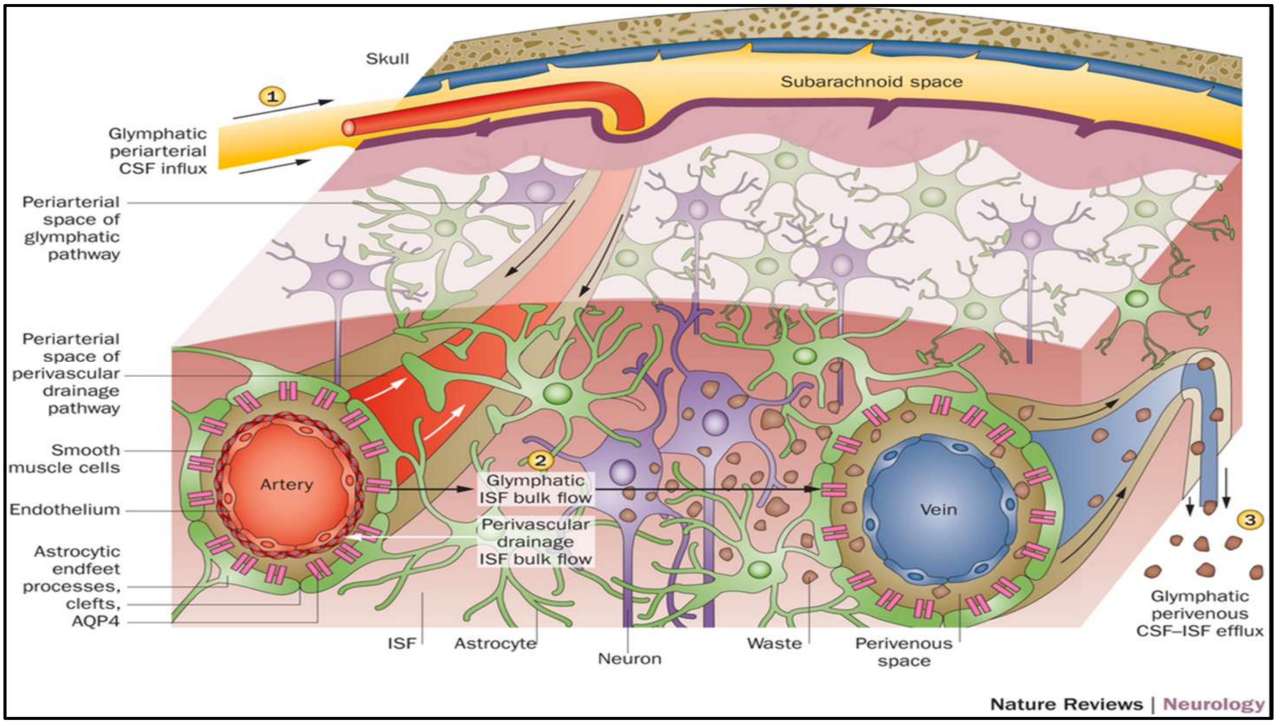
Relatively newly discovered anatomy

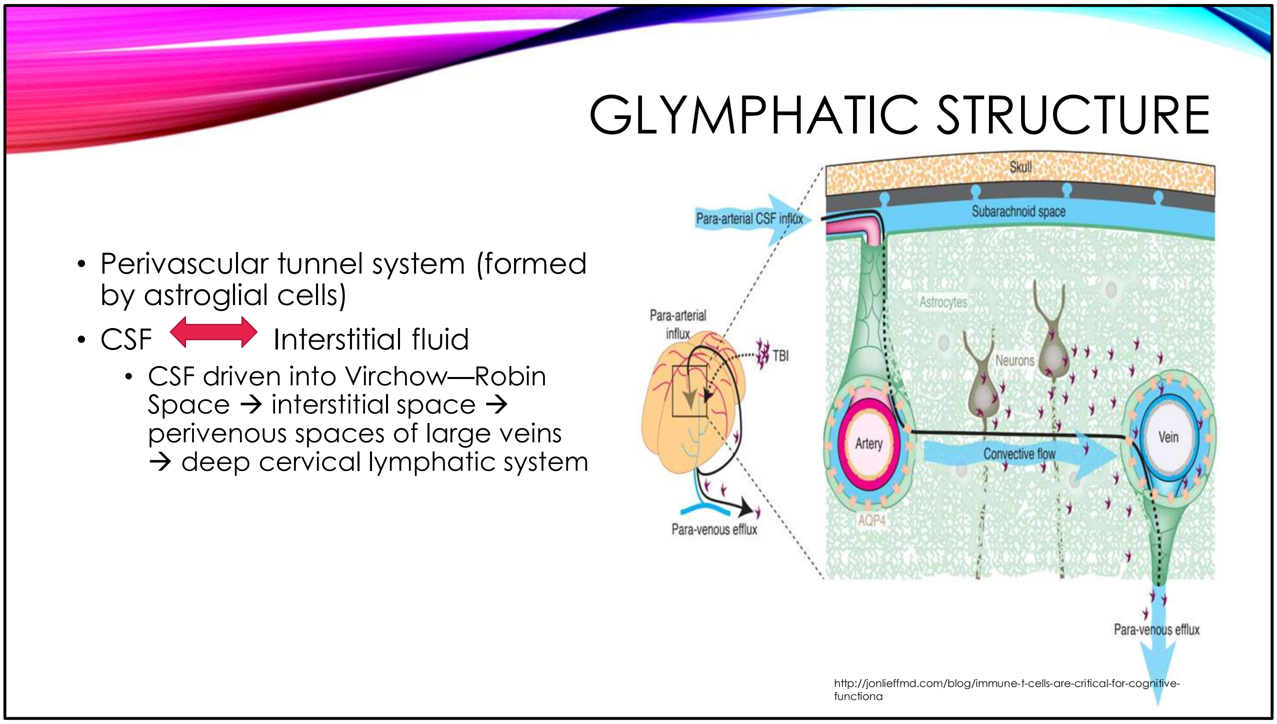
Waste disposal system for the CNS

"Facilitates brain-wide distribution of several compounds, including glucose, lipids, amino acids, growth factors, and neuromodulators"

Functions mainly during sleep

- 60% increase in interstitial space and increased convective flux between CSF and interstitial fluid


Skull
Subarachnoid space
1
Glymphatic periarterial CSF influx
Periarterial space of glymphatic pathway
Periarterial space of perivascular drainage pathway
Smooth muscle cells
Endothelium
Astrocytic endfeet processes, clefts, AQP4
Artery
2
Glymphatic ISF bulk flow
Perivascular drainage ISF bulk flow
Vein
3
Glymphatic perivenous CSF–ISF efflux
ISF
Astrocyte
Neuron
Waste
Perivenous space
Nature Reviews | Neurology



Virchow-robin (space surrounding penetrating arterioles at pia/brain intervace)

LYMPHATIC TECHNIQUES AND TBI

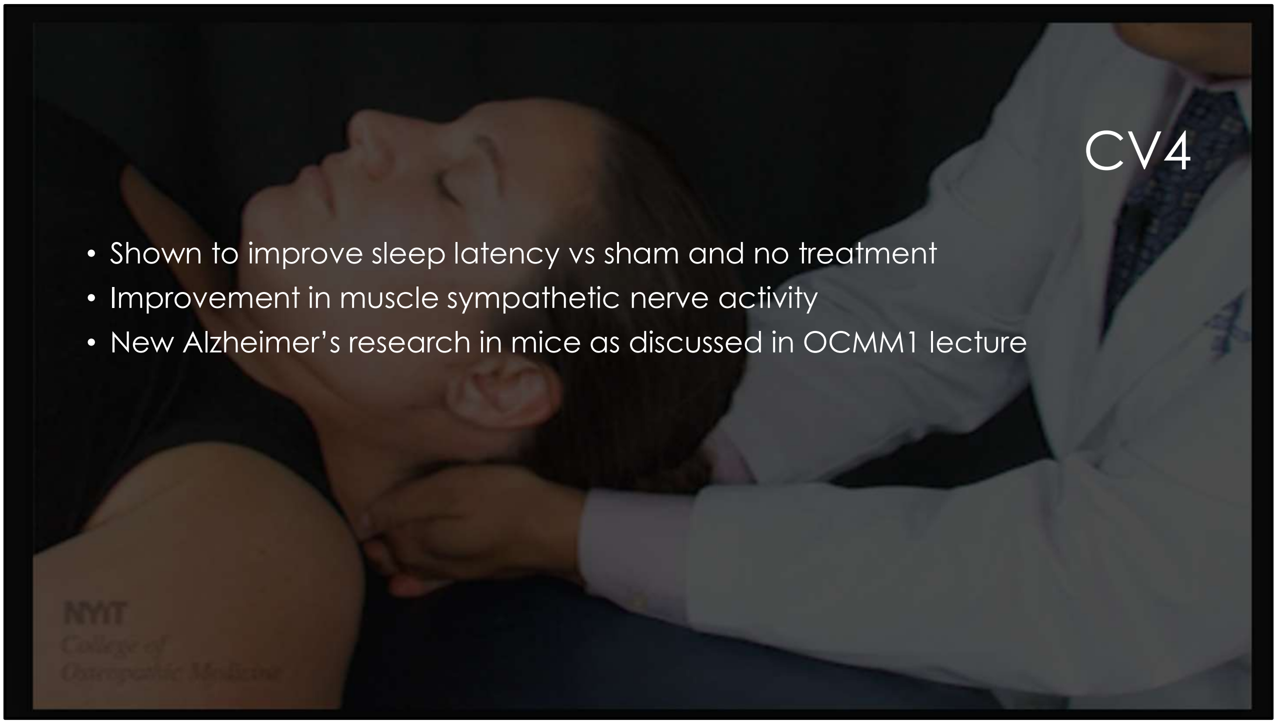
- Performed on 24 comatose patients in ICU
  - 18-69 years old
- Treated with pedal and lymphatic pump
- Group 1: baseline ICP ≤ 20 mmgHg: slight decrease in ICP, increased CCP
- Group 2: baseline ICP >20 mmgHg: decreased ICP, increased CCP

Yale.edu

Monitored at baseline, during OMT, and 5 mintues after
Drains open during OMT except each minute to register ICP
50 treatments with pedal/thoracic pump

LYMPHATIC TECHNIQUES

- With techniques that treat the lymphatic/fluid systems in the body studies have shown:
  - Increased flow toward the thoracic duct
  - Increased leukocyte release into lymph/mobilization of leukocytes
  - Mobilization of inflammatory modulators
  - Improved healing



CV4

- Shown to improve sleep latency vs sham and no treatment
- Improvement in muscle sympathetic nerve activity
- New Alzheimer's research in mice as discussed in OCMM1 lecture

PTSD AND HEAD INJURY

- Received "light touch manual therapy" – craniosacral therapy and brain curriculum
- Done on active-duty soldiers
- 10 men, 27-45 y/o
  - 80% TBI, 90% HA, 100% PTSD
- 2 one-hour sessions one week apart
- Showed decrease in intensity of HA and anxiety, increased cognitive function and mobility

CASE STUDIES

- 16 y/o F with 3 prior head injuries including most recent ~1 month prior
  - Sxs included amnesia, fatigue, HA, labile mood, vertigo, photo and phonophobia, concentration problems, insomnia, and vision problems
  - Resolution of symptoms over ~1 month of OMT
- 27 y/o with TBI after snowboarding
  - Had dizziness, nausea, fogginess, tinnitus
  - Resolution of symptoms following 25 minutes of OMT and improvement in sensory organization test following OMT
- 38 y/o F with 3 months of post-concussion symptoms
  - Multi-disciplinary approach including OMM, assessing psychological factors, pharmacologic interventions, injections, PT

OSTEOPATHIC RESEARCH

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